

The growth series of the generic right-triangle reflection group is not D -finite

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Abstract

Let $W_{\text{univ}} = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the sides of a right triangle with generic angle parameter, to which every right triangle group agrees up to a finite, effectively computable depth [3]. Let $U(x) = \sum_n u_n x^n$ be its growth series with respect to the three side reflections (the integer sequence OEIS A396406 [26]), and V its relaxed companion. We prove that U and V continue meromorphically to the disc $|x| < 1$ and have infinitely many poles there, accumulating at the unit circle. Consequently neither is algebraic over $\mathbb{Q}(x)$ or D -finite, so (u_n) and (v_n) are not P-recursive; neither satisfies a linear q -difference equation with Q not a root of unity or a Mahler equation; and, by the Pólya–Bertrandias theorem, $|x| = 1$ is a natural boundary of both. The poles lie over the zeros of the Hahn–Exton q -Bessel function $J_{-1/2}^{(3)}(Z(q); q^2)$, $Z = \sqrt{2(1-q)/q}$, $q = x^2$, which up to a positive factor is the Fredholm determinant of the travel transfer operator; that there are infinitely many of them follows from a spectral argument with no asymptotic estimate. They survive in U and V because the leading Laurent coefficient at each of them is the square of a junction pairing, which has an exact closed form in one bulk quantity and is positive at every travel pole. For the travel resolvent $\Sigma_0/(1 - \Sigma_1)$ all of this is unconditional. For U and V it rests on the transfer model, a theorem whose assembly step is a hand proof, and on the junction-pairing theorem, whose positivity uses an analytic gate estimate proved partly by verified interval arithmetic. The long proofs are given in a companion paper.

1 Introduction

Growth series of groups. For a group G with a finite generating set S , the growth series $\sum_{n \geq 0} a_n x^n$ counts the elements of G by their word length in S . In the classical families it is rational: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [8]), and Coxeter systems are governed by Steinberg’s rational formula [15, §5.12]. For wreath products, including lamplighter-type groups, Parry computed the growth series for natural generating sets and found them *algebraic* [23]. Provably transcendental growth series are rarer. Stoll [29] showed that the higher Heisenberg group H_5 has a transcendental growth series with respect to its standard generating set, and a rational one with respect to a specially chosen generating set; the three-dimensional Heisenberg group, by contrast, has rational growth for every finite generating set (Duchin–Shapiro [11]). Two general mechanisms give growth series that are not even D -finite. For groups of polynomial growth, and for groups of intermediate growth such as Grigorchuk’s [13], the growth series has radius of convergence 1 and integer coefficients, so by Pólya–Carlson [10] it

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is rational or has $|x| = 1$ as a natural boundary; in the second case, which includes Stoll's series for H_5 and every group of intermediate growth, it is not D -finite. And a finitely presented group with unsolvable word problem has a non-computable growth function, while a D -finite series with rational coefficients has computable coefficients, so its growth series is not D -finite either.

The group. A right triangle T has three sides; the reflections across them generate a group $W_T = \langle R_0, R_1, R_2 \rangle \leq \text{Aff}(\mathbb{R}^2)$. For a right triangle whose angle parameter is generic, W_T is the *universal* group W_{univ} of [3], and every other right triangle group agrees with W_{univ} in the ball of some finite radius and deviates beyond it; the radius is effective [3, Theorem “Effective universality”, thm:effective-universality] (for the $(1, 2)$ triangle the first deviation is at depth 33, [3, Theorem “Sharp uniform universality at depth 32”, thm:universality-sharp]). Irrationality of the angles alone (Niven [21]) does not give genericity: a triangle in $\mathbb{Q}^2$ with unequal legs has irrational angles, yet it is not universal. Throughout, $U(x) = \sum_n u_n x^n$ is the growth series of W_{univ} with respect to $\{R_0, R_1, R_2\}$. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \dots$ is OEIS A396406; it coincides with the Fibonacci numbers F_{n+3} for $1 \leq n \leq 9$ and deviates at $n = 10$. W_{univ} is not a Coxeter group [16], and it is virtually the lamplighter $\mathbb{Z} \wr \mathbb{Z}$ [3]. Its generating set is not constructed but imposed by Euclidean geometry. For a group generated by reflections, Steinberg's formula would make the growth series rational in the Coxeter case, and Parry's theorem makes the natural wreath-product series algebraic; the result below places this geometric system outside both classes.

The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\geq 0}$ a cycle count. The relaxed counting function is the companion V ; one has $v_n \geq u_n$, with equality for $n \leq 7$ and strict inequality at $n = 8$.

The main theorem.

Main Theorem. (i) U and V continue to single-valued meromorphic functions on the disc $|x| < 1$, and each has infinitely many poles there, accumulating at the unit circle (Lemma 13, Theorems 18 and 20).

(ii) Consequently U and V are transcendental over $\mathbb{Q}(x)$ and not D -finite, so (u_n) and (v_n) are not P -recursive (Theorem 26); neither satisfies a linear q -difference equation $\sum_i a_i(x)f(Q^i x) = b(x)$ with Q not a root of unity, nor a Mahler equation (Proposition 28); and $|x| = 1$ is a natural boundary of both (Theorem 30).

(iii) The travel resolvent $G_0^T = \Sigma_0/(1 - \Sigma_1) \in \mathbb{Z}[[q]]$ has all of these properties in the variable q , with natural boundary $|q| = 1$ (Theorems 17, 26, 30, Proposition 28).

(iv) The exponential growth rate of (u_n) and of (v_n) is $\beta_2 = 1.4916177871\dots$, where β_2^{-2} is the smallest zero in $(0, 1)$ of $q \mapsto J_{-1/2}^{(3)}(Z(q); q^2)$ (Corollary 19).

(Standing: (iii) is unconditional. (i), (ii) and (iv) rest on Theorems 12 and 15, as stated there.)

The mechanism. The proof does not go through the coefficients. It locates infinitely many poles of U and V inside the disc $|x| < 1$ of meromorphy, beyond the radius of convergence $1/\beta_2 = 0.6704\dots$, and then applies three classical facts: an algebraic function has finitely many poles in the disc; a pole of a solution of a linear differential equation whose coefficients are holomorphic near the closed disc is a zero of its leading coefficient (Lemma 25); and an integer power series that is meromorphic on a simply connected domain of conformal radius larger than 1 is rational (Pólya–Bertrandias, Theorem 29). The poles are produced in three steps.

(1) *The transfer model.* Grading by relaxed length and marking cycles gives a bivariate series $\mathcal{W}(x, y)$ with $V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, x^2)$. Theorem 12 writes $\mathcal{W}$ as a sum of three sectors. Two of them are pairings against the resolvent $(I - \mathcal{T})^{-1}$ of the *travel operator* $\mathcal{T} = D_e K$, with $q = x^2$, $K = [q^{\max(s, s')}]_{s, s' \geq 0}$ and $D_e = \text{diag}(2q^{1+s})$; the bulk enters only through explicit vectors built from the resolvent of the same semiseparable kernel on a shifted index set, with a rank-one gap term, and the third sector contains no travel resolvent. Lemma 13 continues $\mathcal{W}$ meromorphically to $|x| < 1$.

(2) *The travel poles.* The poles of $(I - \mathcal{T})^{-1}$ are the zeros of its Fredholm determinant, and Theorem 3 computes that determinant in closed form:

$$\det(I - \lambda \mathcal{T}) = \sum_{k \geq 0} \frac{(-2\lambda(1 - q))^k q^{k^2}}{(q; q)_{2k}} = \cos(\sqrt{\lambda} Z; q^2) = c_q(\sqrt{\lambda} Z)^{1/2} J_{-1/2}^{(3)}(\sqrt{\lambda} Z; q^2),$$

the q -cosine of Koornwinder and Swarttouw [18] and the Hahn–Exton q -Bessel function [14, 12, 17]. So the travel poles are the $q \in (0, 1)$ with $J_{-1/2}^{(3)}(Z(q); q^2) = 0$ (Corollary 5). A symmetrisation of $\mathcal{T}$ is a positive trace-class operator with simple eigenvalues $1/\lambda_k(q)$; the branches satisfy $\lambda_1(q) > (1 - q^2)/(2q)$ and $\lambda_k(q)/(1 - q) \rightarrow (2k - 1)^2 \pi^2/8$, so each crosses the level 1, and distinct branches cross at distinct points: there are infinitely many travel poles (Proposition 6), locally finite in $(0, 1)$ (Lemma 7), hence accumulating at $q = 1$. This step uses no asymptotic estimate of a q -series.

(3) *The residue.* At a travel pole q_m the leading Laurent coefficient of $\mathcal{W}(\cdot, y)$ is $c_\star \Pi_y(q_m)^2/(2x_m)$, where $\Pi_y = \langle \lambda^{(y)}, R \rangle$ pairs the junction vector with the null vector R of $I - \mathcal{T}(q_m)$ (Proposition 14). The junction is not of rank one, so the numerator does not factor through the total bulk sum; nevertheless two shift identities for R collapse the pairing to an exact closed form in the single bulk quantity t_1 (Theorem 15), for instance

$$(1 - \mathfrak{g}_V t_1) \Pi_1(q_m) = \frac{2q(1 + x)}{1 - q} \left(t_1 + \frac{1 + x}{2x} \right),$$

and this is positive at every travel pole: analytically for $\tau_m \leq 5 \cdot 10^{-3}$ once the *gate* $|\mathfrak{g}_V t_1| < 1$ holds, and at the first twelve poles by an interval certificate. For the travel resolvent the corresponding non-vanishing is an exact identity, $\Sigma_0(q_m) S_0(q_m) = 2q_m/(1 - q_m)$ (Proposition 11), a consequence of the q -Pythagorean identity at a zero of the q -cosine.

Standing, and what is cited from the companion paper. The statements about the travel resolvent are unconditional. The statements about U and V rest on two theorems. The transfer model (M), Theorem 12: its metric identity and the faithfulness of the model are formalised in Lean 4, and its block decomposition (M3') is a hand proof, given in the companion paper [5], checked against exact enumeration to x^{26} and not formalised. The junction pairing (R-J), Theorem 15: its closed form is proved here; its positivity is analytic for $\tau_m \leq 5 \cdot 10^{-3}$, given the gate, and rests on an interval-arithmetic certificate at the first twelve poles; only the analytic range, which contains all but finitely many travel poles, is needed for (i)–(ii) of the Main Theorem. The gate itself, the amplitude estimate ($\star$) and the lower bound on the bulk denominator at the travel poles, is proved in [5], with constants produced by committed certificates and re-certified in directed-rounding interval arithmetic on the range stated there.

This paper states and proves the spectral and q -trigonometric steps, the local form, the closed form of the junction pairing and every consequence above, and it states the transfer model with an outline of its proof. The long technical proofs are in [5] and are cited by theorem label: the block assembly (the section identity, the telescoping of the travel and bulk sums, the reciprocity and the Möbius factorisation of the gap-marked bulk resolvent), the site cost of the crossing optimisation and

its Lean formalisation, the gap-free theorem behind Proposition 1, the proof of (M3'), the gate with its certificates, the explicit lower bound for S_e at the travel poles, an absolute bound on an auxiliary contour integral T_2 which gives a second proof of the infinitude of the travel poles and localises them in windows of w (the window statement is used only in the Mahler part of Proposition 28), the Pólya–Carlson dichotomy for the four building blocks, the spectral sign laws with their standing, and the value-transcendence question for the number β_2 . The archival full version [4] contains both papers; labels are shared with it.

Related results. The cogrowth series is settled by general theory: W_{univ} is amenable, being virtually metabelian, and not virtually nilpotent, so its cogrowth series is not D -finite by Bell–Mishna [1]. In higher dimensions the picture is subtler [6]: the abstract right-angled Coxeter envelope of an n -orthoscheme has rational growth, but the geometric group is an amenable proper quotient that matches this envelope only to a finite depth. Whether U and V are D -algebraic is open.

Organisation. Section 2 introduces the building blocks and the travel poles. Section 3 identifies the pole equation with the q -cosine and with the Fredholm determinant of the travel operator, proves that the travel poles are infinite and accumulate only at $q = 1$, and proves that the numerators do not vanish there. Section 4 states the transfer model, continues $\mathcal{W}$ meromorphically and computes the shape of its local form at a travel pole; Section 4.3 evaluates the junction pairing. Section 5 proves transcendence, and Section 6 the statements beyond algebraicity. Section 7 lists open questions.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q -difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q ,

$$\Sigma_1(q) = \sum_{i \geq 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \eta_i, \quad S_e(q) = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}, \quad (1)$$

and their companions Σ_0, S_0 , where $\eta_j = e^{-B_j}$ is the Gaussian-decaying form factor of [5, §sec:sd] (equivalently $\eta_j = q^{j^2+j} [2(1-q)]^j (2j)! / [(q; q)_{2j} W^{2j}]$ with $W = w e^{-\tau/2}$; the identification of this display with $1 - \Sigma_1 = \cos(Z; q^2)$ of Proposition 2 is the collapse used in [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles]). We write $\Sigma_1^T = \Sigma_1$ when the travel rôle is to be emphasised. The relaxed travel resolvent is $G_0^T = \Sigma_0 / (1 - \Sigma_1)$ ([5, Proposition “the travel k -sum telescopes”, prop:travelsum]), and U inherits its denominator $1 - \Sigma_1$ (Proposition 1); how the two blocks assemble into V and U , and how far that assembly is proved, is [5, §sec:assembly]. The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \quad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$ (Proposition 6 with Lemma 7; a second proof, through an absolute bound on an auxiliary contour integral, is [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles]).

The four blocks are the k -recursion sums

$$\Sigma_k = \sum_{j \geq 0} A_{k+2j} \prod_{i < j} C_{k+2i}, \quad S_k = \sum_{j \geq 0} \alpha_{k+2j} \prod_{i < j} \gamma_{k+2i}, \quad (2)$$

$$A_k = \frac{2q}{1 - q^{k+1}}, \quad C_k = \frac{2q^{k+3}}{1 - q^{k+2}} - \frac{2q^{k+2}}{1 - q^{k+1}}, \quad \alpha_k = \frac{2q^{k+1}}{1 - q^{k+1}}, \quad \gamma_k = \alpha_{k+1} - \alpha_k,$$

Σ_0, Σ_1 carrying the travel rôle and S_0, S_1 the bulk rôle; Σ_1 and S_1 are the denominators of (1) and Σ_0, S_0 their numerators. Every factor is a Lambert series with integer coefficients, and the j -th summand of Σ_1 starts at order $(j+1)^2$, so each coefficient is a finite integer sum; $\Sigma_1 = 2q + 2q^3 - 4q^4 + 6q^5 - \dots$. All four blocks are holomorphic on $|q| < 1$: by Proposition 2 and Lemma 9 below each is a q -trigonometric series whose j -th term carries the Gaussian factor $q^{j^2+O(j)}$ against a denominator $(q; q)_{2j}$ or $(q; q)_{2j+1}$, bounded below by $|(q; q)_\infty| > 0$ on compact subsets of the disc (this is the first clause of [5, Theorem thm:blocks], where the Pólya–Carlson dichotomy for the four blocks is proved). The coefficients of G_0^T are the travel-run counts 2, 6, 10, 26, 54, 114, 274, $\dots$, integers.

Proposition 1 (Invariance of the travel block). *The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V ; hence U and V share the travel poles q_m .*

Proof. A pure-travel element has no gap edge, since its span is I_k and $f_j = \pm 1$ there (I_k the travel interval and f_j the travel indicator of [5, §sec:sitecost]; this is [5, Corollary cor:localzero]), so [5, Theorem “no gap edge forces $c = 0$ ”, thm:nogap] gives $c = 0$: some relaxed-optimal realisation has no isolated cycle. That realisation is a single open walk, hence admissible for ℓ_T , so $\ell_T \leq \ell_R$; and $\ell_T \geq \ell_R$ because the relaxed minimum ranges over a superset of the realisations. Therefore $\ell_T = \ell_R$ on pure-travel elements and the two gradings of T agree termwise. The travel recursion is assembled from pure-travel runs alone, appending one $f = \pm 1$ edge at a time, so it is computed in the sector $c \equiv 0$ and the travel block carries y^0 only. $\square$

The history of this proposition, which earlier versions carried as the hypothesis (T), is recorded in [5, §sec:inputT].

3 The travel poles

3.1 The travel-pole equation as a q -trigonometric identity

The two series that govern the poles admit an exact identification with the q -cosine of Koornwinder and Swarttouw [18], $\cos(z; q^2) := {}_1\phi_1(0; q; q^2, q^2 z^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j} / (q; q)_{2j}$.

Proposition 2. *Let $z_0 = \sqrt{2(1-q)}$ and $Z = z_0/\sqrt{q}$. Then*

$$1 - S_1 = S_e = \cos(z_0; q^2), \quad 1 - \Sigma_1 = \cos(Z; q^2).$$

In particular the travel poles q_m are exactly the solutions of $\cos(Z; q^2) = 0$, and S_e is the same function evaluated one half q -step below its own zero.

Proof. Collapse the k -recursion (2) on its two odd ladders, the travel one carrying Σ_1 and the bulk one carrying S_1 . Write $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$ in the sinh-forms of the two series ([5, §sec:sd]): the products collapse to $1 - \Sigma_1 = \sum_j (-1)^j (2(1-q))^j q^{j^2} / (q; q)_{2j}$ and $1 - S_1 = \sum_j (-1)^j (2(1-q))^j q^{j^2+j} / (q; q)_{2j}$, the second being S_e as defined in (1). Comparing with the series for $\cos(z; q^2)$ term by term forces $q^2 z^2 = 2(1-q)q$, i.e. $z = Z$, in the first case and $q^2 z^2 = 2(1-q)q^2$, i.e. $z = z_0$, in the second; both series converge for $|q| < 1$, the factor q^{j^2} dominating. Both collapses are recomputed as identities of integer power series, to degree 1160 in `blocks_growth.py` and at two truncation degrees in `f2_pole_route.py`. The per-term identity is machine-checked in Lean 4 (`GramCasoratian.lean`, theorem `sigma1_term`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry). $\square$

The Hahn–Exton normalisation. Write $J_\nu^{(3)}(x; q^2) = x^\nu \frac{(q^{2\nu+2}; q^2)_\infty}{(q^2; q^2)_\infty} {}_1\phi_1(0; q^{2\nu+2}; q^2, q^2 x^2)$ for the Hahn–Exton q -Bessel function, in the normalisation of [17, eq. (2.4)] with base q^2 , and put

$$c_q := \frac{(q^2; q^2)_\infty}{(q; q^2)_\infty} > 0.$$

Since $(q^2; q^2)_k (q; q^2)_k = (q; q)_{2k}$ and $(q^2; q^2)_k (q^3; q^2)_k = (q; q)_{2k+1} / (1 - q)$, comparing coefficients gives, for $x > 0$,

$$\cos(x; q^2) = c_q x^{1/2} J_{-1/2}^{(3)}(x; q^2), \quad \sin(x; q^2) = c_q x^{1/2} J_{1/2}^{(3)}(x; q^2).$$

So the travel poles are the zeros in $(0, 1)$ of $q \mapsto J_{-1/2}^{(3)}(Z(q); q^2)$ (Corollary 5); [5, Proposition “structure of the zero curve”, prop:zerostructure(i)] records the same identity in the variable of [5, Appendix app:beta2].

3.2 The pole equation as a Fredholm determinant

The q -cosine of Proposition 2 is the Fredholm determinant of the travel operator. Throughout this subsection $0 < q < 1$, $D_e = \text{diag}(2q^{1+s})_{s \geq 0}$ and $K = [q^{\max(s, s')}]_{s, s' \geq 0}$, so that $\mathcal{T} = D_e K$ is the travel operator of [5, Definition “section identity”, def:model]. Put

$$\mathbf{S} := D_e^{1/2} K D_e^{1/2}, \quad \mathbf{S}[s, s'] = 2q^{1+(s+s')/2+\max(s, s')}, \quad F(\lambda, q) := \sum_{k \geq 0} \frac{(-2\lambda(1-q))^k q^{k^2}}{(q; q)_{2k}} \quad (\lambda \in \mathbb{C}).$$

The spectral parameter λ is not to be confused with the junction vectors $\lambda^{(y)}$ of §4.

Theorem 3 (the travel determinant). *Let $0 < q < 1$ and $Z = \sqrt{2(1-q)}/q$.*

- (i) $\mathbf{S}$ is a positive definite trace-class operator on $\ell^2(\mathbb{Z}_{\geq 0})$, with $\text{tr } \mathbf{S} = 2q/(1 - q^2)$.
- (ii) $\mathcal{T}$ and $\mathbf{S}$ have the same principal minors, $\det \mathcal{T}[I, I] = \det \mathbf{S}[I, I]$ for every finite $I \subset \mathbb{Z}_{\geq 0}$, and the same non-zero eigenvalues, the eigenvectors corresponding by $R = D_e^{1/2} \phi$: if $\mathbf{S} \phi = \kappa \phi$ with $\phi \in \ell^2$ and $\kappa \neq 0$, then $R \in \ell^1$ and $\mathcal{T} R = \kappa R$; conversely, if $R \in \ell^1$ and $\mathcal{T} R = \kappa R$ with $\kappa \neq 0$, then $\phi = D_e^{-1/2} R \in \ell^2$ and $\mathbf{S} \phi = \kappa \phi$.
- (iii) For every $\lambda \in \mathbb{C}$, with absolutely convergent series,

$$\det(I - \lambda \mathbf{S}) = \sum_{k \geq 0} (-\lambda)^k \sum_{|I|=k} \det \mathbf{S}[I, I] = F(\lambda, q),$$

$$F(\lambda, q) = \cos(\sqrt{\lambda} Z; q^2) = c_q (\sqrt{\lambda} Z)^{1/2} J_{-1/2}^{(3)}(\sqrt{\lambda} Z; q^2),$$

the last two expressions being even in $\sqrt{\lambda}$. We write $\det(I - \lambda \mathcal{T})$ for the same Fredholm series formed with the minors of $\mathcal{T}$; by (ii) it is the same function.

- (iv) The eigenvalues of $\mathbf{S}$ are $1/\lambda_k(q)$, $k \geq 1$, with $0 < \lambda_1(q) < \lambda_2(q) < \dots \rightarrow \infty$, all simple; $\sum_k 1/\lambda_k(q) = 2q/(1 - q^2)$ and $F(\lambda, q) = \prod_{k \geq 1} (1 - \lambda/\lambda_k(q))$.
- (v) $1 - \Sigma_1 = F(1, q) = \det(I - \mathcal{T})$, and $S_e = F(q, q) = \det(I - q\mathcal{T}) = \det(I - M_0)$: the bulk operator M_0 of [5, Definition “section identity”, def:model] is $q\mathcal{T}$ after the index shift $b = s + 1$.

Proof. (i) For $x \in \ell^2$ put $X = D_e^{1/2}x$, which lies in ℓ^1 by Cauchy–Schwarz. Since $q^{\max(s,s')} = (1-q) \sum_{t \geq \max(s,s')} q^t$,

$$\langle x, \mathbf{S}x \rangle = \langle X, KX \rangle = (1-q) \sum_{t \geq 0} q^t \left| \sum_{s \leq t} X_s \right|^2 \geq 0,$$

the double sums converging absolutely, with equality only if every partial sum of X vanishes, that is $X = 0$, that is $x = 0$. $\mathbf{S}$ is Hilbert–Schmidt, hence bounded, and it is positive definite; a positive operator with summable diagonal is trace class, with trace $\sum_s 2q^{1+s}q^s = 2q/(1-q^2)$.

(ii) $\mathcal{T}[I, I] = D_e[I, I]K[I, I]$ and $\mathbf{S}[I, I] = D_e[I, I]^{1/2}K[I, I]D_e[I, I]^{1/2}$ with $D_e[I, I]$ diagonal, so the two determinants agree. If $\mathbf{S}\phi = \kappa\phi$, then $R = D_e^{1/2}\phi \in \ell^1$ and $\mathcal{T}R = D_e^{1/2}\mathbf{S}\phi = \kappa R$. Conversely, $\mathcal{T}R = \kappa R$ gives $R = \kappa^{-1}D_e(KR)$ with $|(KR)_s| \leq \|R\|_{\ell^1}$, so $\phi = D_e^{-1/2}R = \kappa^{-1}D_e^{1/2}KR \in \ell^2$ and $\mathbf{S}\phi = D_e^{1/2}KR = \kappa\phi$. The operators are not boundedly similar, $D_e^{-1/2}$ being unbounded; only (ii) is used.

(iii) For trace-class A , $\det(I - \lambda A) = \sum_k (-\lambda)^k \text{tr } \Lambda^k(A)$ [25, Ch. 3] (the von Koch form; compare Fredholm’s formula for continuous kernels, [25, Thm. 3.10]), and computing the trace of $\Lambda^k(\mathbf{S})$ in the orthonormal basis $e_{s_1} \wedge \cdots \wedge e_{s_k}$, $s_1 < \cdots < s_k$, whose diagonal entries are the principal minors, gives the first equality. The principal minors of the positive $\mathbf{S}$ are ≥ 0 , so the inner sums converge absolutely, to the values computed next, and the outer series then converges absolutely for every λ . For $I = \{s_1 < \cdots < s_k\}$ the matrix $K[I, I]$ is $(a_{\max(i,j)})$ with $a_i = q^{s_i}$ decreasing; subtracting row $i+1$ from row i for $i < k$ leaves a lower triangular matrix with diagonal $a_1 - a_2, \dots, a_{k-1} - a_k, a_k$, so $\det K[I, I] = q^{s_k} \prod_{i < k} (q^{s_i} - q^{s_{i+1}})$ and, with the gaps $g_i = s_{i+1} - s_i \geq 1$,

$$\det \mathbf{S}[I, I] = 2^k q^k q^{2(s_1 + \cdots + s_k)} \prod_{i < k} (1 - q^{g_i}).$$

Since $s_1 + \cdots + s_k = ks_1 + \sum_{j < k} (k-j)g_j$, the sum over $s_1 \geq 0$ and $g_1, \dots, g_{k-1} \geq 1$ factorises; with $\sum_{g \geq 1} q^{2rg}(1-q^g) = q^{2r}(1-q)/((1-q^{2r})(1-q^{2r+1}))$,

$$\sum_{|I|=k} \det \mathbf{S}[I, I] = \frac{(2q)^k}{1-q^{2k}} \prod_{r=1}^{k-1} \frac{q^{2r}(1-q)}{(1-q^{2r})(1-q^{2r+1})} = \frac{(2(1-q))^k q^{k^2}}{(q; q)_{2k}}.$$

Hence $\det(I - \lambda \mathbf{S}) = F(\lambda, q)$; the case $k = 1$ is the trace. Finally $(2\lambda(1-q))^k q^{k^2} = q^{k^2+k}(\sqrt{\lambda} Z)^{2k}$, since $Z^2 = 2(1-q)/q$, so $F(\lambda, q) = \cos(\sqrt{\lambda} Z; q^2)$, and the Hahn–Exton form is the normalisation after Proposition 2.

(iv) For positive trace-class $\mathbf{S}$ the determinant is the product $\prod_k (1 - \lambda \kappa_k)$ over its eigenvalues $\kappa_k > 0$ counted with multiplicity [25, Ch. 3], so the $1/\kappa_k$ are the zeros of $F(\cdot, q)$, the multiplicity of κ_k being the order of the zero. By (iii) these are the $\lambda = x^2/Z^2$ with $x > 0$ a zero of $\cos(x; q^2) = c_q x^{1/2} J_{-1/2}^{(3)}(x; q^2)$ ($\cos(\cdot; q^2)$ is even with $\cos(0; q^2) = 1$). For $\nu > -1$ the zeros of $J_\nu^{(3)}(\cdot; q^2)$ are real and simple and form an infinite sequence [17, Cor. 3.2, Lemma 3.3, Thm. 3.4]; here $\nu = -\frac{1}{2}$ with base q^2 . So the eigenvalues are simple and infinitely many, and the sum of the $1/\lambda_k$ is the trace.

(v) At $\lambda = 1$, $F(1, q) = \sum_k (-1)^k (2(1-q))^k q^{k^2} / (q; q)_{2k} = 1 - \Sigma_1$ (Proposition 2); at $\lambda = q$, $F(q, q) = \sum_k (-2(1-q))^k q^{k^2+k} / (q; q)_{2k} = S_e$ by (1). And $M_0[s+1, s'+1] = 2q^{s+1} q^{\max(s,s')+1} = q \mathcal{T}[s, s']$, so M_0 has the principal minors of $q \mathcal{T}$. $\square$

The identity $\det(I - \lambda \mathbf{S}) = F(\lambda, q)$ was checked in double precision against truncated matrices at $q = 0.3, 0.6, 0.8$ and four values of λ (one of them non-real), to within $3.2 \cdot 10^{-15}$, and the gap sum against a brute-force sum of minors for $k \leq 3$ at $q = 0.3$.

Remark 4 (a q -string). With the Jackson integral $\int_0^1 f d_q t = (1-q) \sum_{s \geq 0} q^s f(q^s)$, let $K_q f(x) = \int_0^1 \min(x, t) f(t) d_q t$ on the lattice $x \in \{q^s\}$. At $x = q^s$, $(KD_e u)_s = \sum_{s'} q^{\max(s, s')} 2q^{1+s'} u_{s'} = \frac{2q}{1-q} (K_q u)(q^s)$, that is,

$$\mathcal{T}^\top = KD_e = \frac{2q}{1-q} K_q,$$

so up to the transposition, which changes neither the principal minors nor the eigenvalues, $\mathcal{T} = \frac{2q}{1-q} K_q$. The kernel $\min(x, t)$ is the Green function of $-d^2/dx^2$ on $[0, 1]$ with a Dirichlet end at 0 and a Neumann end at 1: $\mathcal{T}$ is the compliance operator of a q -string with uniform Jackson mass, fixed at 0 and free at 1, and $u = KR$ is its displacement ($u_s \rightarrow 0$ as $x = q^s \rightarrow 0$). As $q \rightarrow 1$ the kernel tends to the classical one, with eigenvalues $4/((2k-1)^2 \pi^2)$; this is the heuristic behind $\lambda_k(q) \sim (1-q)(2k-1)^2 \pi^2/8$, proved for fixed k in Proposition 6.

Corollary 5 (the travel poles are Hahn–Exton zeros). *A point $q \in (0, 1)$ is a travel pole if and only if*

$$J_{-1/2}^{(3)}(Z(q); q^2) = 0, \quad Z(q) = \sqrt{2(1-q)/q},$$

if and only if 1 is an eigenvalue of $\mathcal{T}(q)$, that is $\lambda_k(q) = 1$ for some k . That k is unique, the eigenvalue is simple, its eigenvector is the travel null vector R of Proposition 14, and $\phi = D_e^{-1/2} R$ is the corresponding eigenvector of $\mathbf{S}$.

Proof. Theorem 3(iii)–(v), with $c_q > 0$ and $Z > 0$, and Proposition 2; uniqueness of k is the simplicity in Theorem 3(iv), and the eigenvector is Theorem 3(ii) with [5, Lemma “the null space is at most one-dimensional”, lem:nullspace]. $\square$

The zero theory used is that of Koelink–Swarttouw [17], at $\nu = -\frac{1}{2}$ and base q^2 , where its hypothesis $\nu > -1$ holds. The eigenvalues also agree with the Jacobi-matrix description of the zeros of $J_\nu^{(3)}$ by Štampach and Štovíček [27], in its Friedrichs extension; the positive form of $\mathbf{S}$ is the Friedrichs realisation, on which $u_s = (KR)_s \rightarrow 0$. At $q_1, \dots, q_{10}$, located at 90-digit precision, $|J_{-1/2}^{(3)}(Z; q^2)| \leq 2 \cdot 10^{-79}$, and exactly $m-1$ zeros of $\cos(\cdot; q_m^2)$ lie below $Z(q_m)$ (a numerical check, not a certificate).

Proposition 6 (infinitely many travel poles, without T_2). (a) $\lambda_1(q) > (1-q^2)/(2q)$ on $(0, 1)$. In particular every $\lambda_k(q) > 1$ for $q \leq \sqrt{2}-1$, and there is no travel pole in $(0, \sqrt{2}-1]$.

(b) For each fixed k , $\lambda_k(q)/(1-q) \rightarrow (2k-1)^2 \pi^2/8$ as $q \rightarrow 1^-$.

(c) Each λ_k is continuous on $(0, 1)$, and there is $c_k \in (\sqrt{2}-1, 1)$ with $\lambda_k(c_k) = 1$. The c_k are pairwise distinct travel poles; in particular there are infinitely many travel poles.

None of this uses T_2 , [5, Lemma “absolute bound on T_2 ”, lem:T2abs], or a single-crossing property.

Proof. (a) The largest eigenvalue of the positive trace-class $\mathbf{S}$ is strictly less than its trace, since $\mathbf{S}$ has infinitely many positive eigenvalues: $1/\lambda_1 < 2q/(1-q^2)$. And $(1-q^2)/(2q) \geq 1$ exactly for $q \leq \sqrt{2}-1$. At such q no λ_k equals 1, so $1 - \Sigma_1 = F(1, q) \neq 0$ by Theorem 3(iv).

(b) Put $\varepsilon = 1-q$ and $\lambda = \varepsilon y$. The k -th term of $F(\varepsilon y, q)$ is $t_k(y, \varepsilon) = (-1)^k (2y\varepsilon^2)^k q^{k^2} / (q; q)_{2k}$. By the arithmetic–geometric mean inequality, $1 - q^n = \varepsilon \sum_{i < n} q^i \geq n\varepsilon q^{(n-1)/2}$, so $(q; q)_{2k} \geq \varepsilon^{2k} (2k)! q^{k(2k-1)/2}$ and

$$|t_k(y, \varepsilon)| \leq \frac{(2|y|)^k q^{k/2}}{(2k)!} \leq \frac{(2|y|)^k}{(2k-1)!!},$$

uniformly in $\varepsilon \in (0, 1)$. Each $t_k(y, \varepsilon)$ tends to $(-1)^k(2y)^k/(2k)!$ as $\varepsilon \rightarrow 0$, because $q^{k^2} \rightarrow 1$ and $(q; q)_{2k}/\varepsilon^{2k} \rightarrow (2k)!$. The domination is summable locally uniformly in $y \in \mathbb{C}$, so $F(\varepsilon y, 1 - \varepsilon) \rightarrow \cos \sqrt{2}y$ uniformly on compact sets. The limit has the simple zeros $y_j = (2j - 1)^2\pi^2/8$, and the zeros of $y \mapsto F(\varepsilon y, 1 - \varepsilon)$ are the $\lambda_j(q)/\varepsilon$, positive and ordered. By Hurwitz's theorem, for ε small each disc $|y - y_j| < \delta$, $j \leq k$, contains exactly one of them and $|y| \leq y_k + \delta$ contains no other; hence $\lambda_j(q)/\varepsilon \rightarrow y_j$ for $j \leq k$.

(c) $q \mapsto \mathbf{S}(q)$ is continuous in trace norm, hence in operator norm, on $(0, 1)$: its entries are continuous and $\sum_{s, s'} |\mathbf{S}[s, s']|$ converges locally uniformly. The ordered eigenvalues of positive compact operators are 1-Lipschitz in operator norm (min-max), so each $1/\lambda_k$, and so each λ_k , is continuous. By (a) $\lambda_k > 1$ on $(0, \sqrt{2} - 1]$, and by (b) $\lambda_k(q) \rightarrow 0$ as $q \rightarrow 1$, so the intermediate value theorem gives c_k ; it is a travel pole by Corollary 5. If $c_j = c_k$ with $j \neq k$, then $\lambda_j(c_k) = \lambda_k(c_k) = 1$, against simplicity (Theorem 3(iv)).

The same argument with $\lambda_k(q)/q$ in place of $\lambda_k(q)$ gives [5, Lemma “infinitude of the zeros of the bulk block”, lem:infzeros] without T_2 too: $S_e = F(q, q) = \prod_k (1 - q/\lambda_k(q))$ vanishes exactly where some $\lambda_k(q) = q$; $\lambda_k(q)/q > (1 - q^2)/(2q^2) > 1$ for $q < 1/\sqrt{3}$ by (a); $\lambda_k(q)/q \rightarrow 0$ by (b); and two branches cannot both equal q at the same point. $\square$

3.3 Local finiteness, and accumulation at $q = 1$

Lemma 7 (The travel poles are locally finite). *For every $\tau_\star > 0$ there are only finitely many travel poles with $\tau_m > \tau_\star$.*

Proof. By Proposition 2 the travel poles are the zeros in $q \in (0, 1)$ of $g(q) := \cos(Z; q^2) = C(Z^2)$ with $Z^2 = 2(1 - q)/q$, i.e. of

$$g(q) = \sum_{j \geq 0} (-1)^j \frac{2^j q^{j^2} (1 - q)^j}{(q; q)_{2j}}.$$

First, g has no zero near $q = 0$. For $0 < q \leq \frac{1}{10}$ one has $(q; q)_{2j} \geq \prod_{i \geq 1} (1 - q^i) \geq 1 - \frac{q}{1 - q} \geq \frac{8}{9}$ and $(1 - q)^j \leq 1$, so $|g(q) - 1| \leq \frac{9}{8} \sum_{j \geq 1} 2^j q^{j^2} \leq \frac{9}{8} (2q + 4q^4/(1 - q)) \leq 0.24 < 1$, whence $g(q) > 0$ there.

Second, g is real-analytic on $(0, 1)$: the series converges locally uniformly, since on $q \leq 1 - \delta$ the j -th term is bounded by $2^j q^{j^2}/(q; q)_\infty$ and q^{j^2} dominates 2^j . Hence the zeros of g are isolated in $(0, 1)$, and any zero set with an accumulation point in the *open* interval would force $g \equiv 0$, contradicting $g(\frac{1}{10}) > 0$.

Now fix $\tau_\star > 0$. The poles with $\tau_m > \tau_\star$ lie in $q \in [\frac{1}{10}, e^{-\tau_\star}]$, a compact subinterval of $(0, 1)$. An analytic function on a domain whose zeros are isolated has only finitely many zeros on any compact subset; hence there are finitely many such poles. $\square$

Remark 8. Only local finiteness is proved here; that the poles *do* accumulate at $q = 1$ (equivalently $\tau_m \downarrow 0$) needs in addition that there are infinitely many of them, which is Proposition 6(c) (see the paragraph after this remark), or, localised in windows of w , [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles]. The present lemma says merely that they cannot accumulate anywhere inside $(0, 1)$, so that any threshold $\tau_\star$ leaves a finite set to be checked directly. Its proof also shows $g > 0$ on $(0, \frac{1}{10}]$, so every travel pole has $q_m > \frac{1}{10}$. Besides the accumulation statement used in Theorems 17, 18 and 20, it makes the splice in [5, Proposition prop:selower] well posed and the exceptional set of [5, Theorem “the amplitude estimate ($\star$)”, thm:star] finite, and it gives the accumulation clause of [5, Lemma lem:infpoles].

With Proposition 6(c), Lemma 7 gives: *the travel poles form an infinite subset of $(\frac{1}{10}, 1)$ whose only accumulation point is $q = 1$; in particular, for every $\tau_\star > 0$ all but finitely many travel poles have*

$\tau_m \leq \tau_*$. This is the form in which the pole set is used below. It needs no estimate on T_2 and no numerical input.

3.4 The numerators, and why they do not vanish at the poles

Proposition 2 identifies the two *denominators*. The two numerators are the companion q -sine at the same two arguments, and that is what settles their non-vanishing at the travel poles. Throughout, $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$ is the half-step q -sine of [5, Lemma “the gate block in closed form”, lem:P12closed], $\sin(z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j+1} / (q; q)_{2j+1}$.

Lemma 9 (the numerators in closed form). *Identically in $0 < q < 1$,*

$$\Sigma_0 = \frac{2q}{Z} \mathfrak{s}(Z), \quad S_0 = \frac{2q}{z_0} \mathfrak{s}(z_0).$$

With Proposition 2 this identifies all four blocks as one q -trigonometric pair sampled at two arguments a half q -step apart: $(\Sigma_0, 1 - \Sigma_1)$ is $(\frac{2q}{u} \mathfrak{s}(u), \cos(u; q^2))$ at $u = Z$, and (S_0, S_e) is the same pair at $u = z_0 = \sqrt{q} Z$. In particular $S_0 = \frac{2q}{1-q} S_o$ with S_o the odd bulk block of [5, Lemma “the gate block in closed form”, lem:P12closed], which is the second identity of the dictionary [5, (eq:dict)] read as a statement about the explicit series, and is therefore independent of the transfer model.

Proof. Collapse the k -recursion (2) on the even ladders, exactly as Proposition 2 does on the odd ones. From $\alpha_k = 2q^{k+1}/(1 - q^{k+1})$,

$$|\gamma_k| = \alpha_k - \alpha_{k+1} = \frac{2(1-q)q^{k+1}}{(1-q^{k+1})(1-q^{k+2})}, \quad |C_k| = \frac{2(1-q)q^{k+2}}{(1-q^{k+1})(1-q^{k+2})},$$

γ_k and C_k themselves being negative on $(0, 1)$, so both even ladders alternate. For the bulk one, $\sum_{i < j} (2i+1) = j^2$ and the denominators telescope to $(q; q)_{2j}$, giving $\prod_{i < j} |\gamma_{2i}| = [2(1-q)]^j q^{j^2} / (q; q)_{2j}$ and hence

$$S_0 = \sum_{j \geq 0} (-1)^j \alpha_{2j} \prod_{i < j} |\gamma_{2i}| = 2 \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{(j+1)^2}}{(q; q)_{2j+1}}.$$

Write $q^{(j+1)^2} = q^{j^2+j} q^j$ and $[2(1-q)]^j q^j = (q^{1/2} z_0)^{2j}$: the sum is $(2q/(q^{1/2} z_0)) \sin(q^{1/2} z_0; q^2) = (2q/z_0) \mathfrak{s}(z_0)$. For the travel ladder, $\sum_{i < j} (2i+2) = j^2 + j$ and the denominators telescope to $(q; q)_{2j}$ again, so $\prod_{i < j} |C_{2i}| = [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$ and, with $A_{2j} = 2q/(1 - q^{2j+1})$,

$$\Sigma_0 = 2q \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{j^2+j}}{(q; q)_{2j+1}} = \frac{2q}{z_0} \sin(z_0; q^2) = \frac{2q^{3/2}}{z_0} \mathfrak{s}(Z) = \frac{2q}{Z} \mathfrak{s}(Z),$$

using $[2(1-q)]^j = z_0^{2j}$ and $z_0 = q^{1/2} Z$. The S_o statement is the identity $S_o = \frac{z_0}{2} \mathfrak{s}(z_0)$ of [5, Lemma “the gate block in closed form”, lem:P12closed] combined with $z_0^2 = 2(1-q)$. Both collapses are the exponent bookkeeping machine-checked in `QTrigIdentities.lean` and are verified numerically off the poles, hence as identities, in `qnumerator_certificate.py`. $\square$

Lemma 10 (half-step rules and the pole invariant). *Identically in $0 < q < 1$, with $c(u) = \cos(u; q^2)$:*

- (i) $c(qu) = c(u) + q^{3/2} u \mathfrak{s}(q^{1/2} u)$;
- (ii) $\mathfrak{s}(qu) = \mathfrak{s}(u) - u c(q^{1/2} u)$;
- (iii) $c(u) c(q^{1/2} u) + q^{3/2} \mathfrak{s}(u) \mathfrak{s}(q^{1/2} u) = 1$.

Consequently, at every travel pole, where $c(Z) = 0$:

$$q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1,$$

so neither $\mathfrak{s}(Z)$ nor $\mathfrak{s}(z_0)$ vanishes there.

Proof. (i) and (ii) are termwise. From $(q; q)_{2j} = (q; q)_{2j-1}(1 - q^{2j})$,

$$c(u) - c(qu) = \sum_{j \geq 1} (-1)^j q^{j^2+j} u^{2j} / (q; q)_{2j-1};$$

shifting $j \mapsto j+1$ and factoring out $-qu$ leaves $-qu \sin(qu; q^2) = -q^{3/2} u \mathfrak{s}(q^{1/2} u)$, which is (i). From $(q; q)_{2j+1} = (q; q)_{2j}(1 - q^{2j+1})$, $\mathfrak{s}(u) - \mathfrak{s}(qu) = q^{-1/2} \sum_{j \geq 0} (-1)^j q^{j^2+j} (q^{1/2} u)^{2j+1} / (q; q)_{2j}$; factoring out $q^{1/2} u$ leaves $u c(q^{1/2} u)$, which is (ii). This is the exponent matching machine-checked as `cshift_exponent` and `sshift_exponent`. For (iii), substitute (i) and (ii) into the Hahn–Exton q -Wronskian $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$ ([30, eq. (4.3.8)]; [22, eq. (4.21)]): the $\mathfrak{s}(u)c(u)$ terms cancel and what is left is $u[c(u)c(q^{1/2} u) + q^{3/2} \mathfrak{s}(u)\mathfrak{s}(q^{1/2} u)]$, so the q -Wronskian and (iii) are equivalent for $u \neq 0$. Identity (iii) is the q -Pythagorean identity of [18]. Taking $u = Z$ and $c(Z) = 0$ gives the pole invariant, which is the step already used inside the proof of [5, Proposition prop:P12second]. $\square$

Proposition 11 (the numerators do not vanish at the travel poles). *At every travel pole q_m ,*

$$\Sigma_0(q_m) S_0(q_m) = \frac{2q_m}{1 - q_m} \neq 0. \quad (3)$$

In particular $\Sigma_0(q_m) \neq 0$ and $S_0(q_m) \neq 0$, at every travel pole, with no asymptotic input and no exceptional set. Moreover $|\Sigma_0(q_m)| |S_0(q_m)| \geq q_m w_m^2$, so at least one of the two numerators exceeds $\sqrt{q_m} w_m$ in modulus.

Proof. By Lemma 9, $\Sigma_0 S_0 = (4q^2/(z_0 Z)) \mathfrak{s}(Z) \mathfrak{s}(z_0)$, and by Lemma 10 the product $\mathfrak{s}(Z) \mathfrak{s}(z_0)$ equals $q^{-3/2}$ at a travel pole. Since $z_0 Z = z_0^2 / \sqrt{q} = 2(1 - q) / \sqrt{q}$, the two powers of q combine to $4q^2 \cdot q^{-3/2} \cdot \sqrt{q} / (2(1 - q)) = 2q / (1 - q)$. The size statement follows from $1 - q \leq \tau = 2/w^2$. $\square$

4 The transfer model and the local form at a travel pole

This section connects the explicit operators to the growth series. We first collect the objects the transfer model is phrased in; their construction, and every identity between them that is used below, is proved in [5, §sec:assembly].

4.1 Notation

Throughout, $|q| < 1$, $x = q^{1/2}$, and at a travel pole q_m we write $\tau_m = -\log q_m$, $w_m = \sqrt{2/\tau_m}$ and $x_m = \sqrt{q_m}$.

The two operators. On $\ell^1(\mathbb{Z}_{\geq 0})$ the *travel operator* is $\mathcal{T} = D_e K$ with $K = [q^{\max(s, s')}]_{s, s' \geq 0}$ and $D_e = \text{diag}(e_s)$, $e_s = 2q^{1+s}$; on $\ell^1(\mathbb{Z}_{\geq 1})$ the *bulk operator* is $M_0[b, a] = 2q^b q^{\max(a, b)}$. These are the travel and bulk instances of the section identity of [5, Definition “section identity”, def:model]; both are compact, and $(I - \mathcal{T})^{-1}$ and $R_0 = (I - M_0)^{-1}$ are meromorphic on $|q| < 1$ with poles only where $1 - \Sigma_1 = 0$, respectively $S_e = 0$ [5, Corollary “singularities of the two gapless resolvents”, cor:sing]. Put $E_b = 2q^b$, $u_b = 2q^{2b}$, $v_a = q^a$ and

$$t_1 = v^\top R_0 u,$$

an explicit bulk quantity; one has $t_1 = P_{12}/S_e$ identically, where P_{12} is the off-diagonal entry of the bulk cocycle, a Hahn–Exton function $J_{3/2}^{(3)}$ in closed form [5, Corollary “the cocycle is the same object”, cor:cocycle].

The gap term. With $\mathfrak{g} = \mathfrak{g}(q, y) = \sum_{L \geq 1} q^L y^{L-1} = q/(1 - qy)$, the gap-marked bulk system is

$$K_{\mathfrak{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathfrak{g}, \quad P_b = 2q^b \left(q^b + \sum_{a \geq 1} K_{\mathfrak{g}}(a, b) P_a \right) \quad (b \geq 1). \quad (4)$$

The gap term is of rank one, $2q^b \cdot q^{a+b} = u_b v_a$, and by the Sherman–Morrison formula $P = B_0 R_0 u$ with

$$B_0 = 1 + \mathfrak{g} \sum_{b \geq 1} q^b P_b = \frac{1}{1 - \mathfrak{g} t_1}, \quad B_{0z} = 1 + y \mathfrak{g} \sum_{b \geq 1} q^b P_b$$

[5, Proposition “Möbius factorisation”, prop:mobius; Remark “the bulk sum of the model”, rem:bulksources]. The two evaluations of the gap kernel are $\mathfrak{g}_V = \mathfrak{g}(q, 1) = \frac{q}{1-q}$ and $\mathfrak{g}_U = \mathfrak{g}(q, q) = \frac{q}{1-q^2}$.

The junction vectors. For $s \geq 0$,

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{b \geq 1} \frac{P_b}{2} \left(x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)} \right), \quad (5)$$

$$E_s^{(y)} = B_0^{[s]} x^{2s} + B_0 x^{2s+2} + \sum_{b \geq 1} P_b \left(x^{\max(2s, 2b)} + x^{\max(2s+2, 2b)} \right), \quad B_0^{[0]} = B_{0z}, \quad B_0^{[s]} = B_0 \quad (s \geq 1), \quad (6)$$

and $\lambda^{(y)} = E^{(y)} + 2\mu^{(y)}$. These are the near- and far-junction vectors of the marker sites; the site costs they encode are computed in [5, Corollary “the marker junctions”, cor:marker], and the rank of the junction is discussed in [5, Proposition “the marker junction is not rank one”, prop:junction].

The bivariate series. With $c(g)$ the cycle count and $\ell_R(g)$ the relaxed length,

$$\mathcal{W}(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c, \quad w_{n, c} = \#\{g : \ell_R(g) = n, \quad c(g) = c\},$$

so that $V(x) = \mathcal{W}(x, 1)$ and, since $\ell_T = \ell_R + 2c$ for every element (formalised in Lean, `metricAll`), $U(x) = \mathcal{W}(x, x^2) = \mathcal{W}(x, q)$ [5, §sec:bridge].

The gate. Write $s = \mathfrak{g}_V t_1 = \frac{q}{1-q} t_1$ and $b_0 = S_0/(1 - S_1)$. The gate is

$$b_0 > 0 \quad \text{and} \quad s < 1; \quad (7)$$

it holds at every travel pole, with $|s| \leq 0.983$ for $\tau_m \leq 5 \cdot 10^{-3}$ and by direct evaluation at the six poles above that threshold, whose enumeration has the standing stated in [5, Theorem “the amplitude estimate ($\star$)”, thm:star] (see [5, Corollary “the gate closes”, app:gate]). The half $|s| < 1$ holds unconditionally, and the half $b_0 > 0$ through the closed form of b_0 , which rests on the transfer model.

4.2 The model (M)

The strand-walk input that identifies the operators above with the growth series is stated once and named.

Theorem 12 (the transfer model (M)). (M1) Local cost law. *The site cost between adjacent active edges carrying deposits d, d' is $x^{\max(|d|, |d'|)}$, independent of the crossing counts and of the travel indicator. Bulk deposits are even, $|d| = 2s$, and travel deposits odd, $|d| = 2s + 1$, so this cost is $q^{\max(s, s')}$ in the bulk and $x^{2\max(s, s') + 1}$ on the travel side, and the edge weights are $2q^s$ and $2x^{2s+1}$ respectively: [5, Definition “section identity”, def:model] in its two instances. At the near marker site the cost is $x^{\text{Site}_0(d_L, d_R)}$ with $\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|)$, d_L the last bulk deposit and d_R the first travel deposit, for all four marker data $(\varepsilon^*, \delta^*)$; at the far marker site it is the $(\varepsilon^*, \delta^*)$ -dependent cost of [5, Corollary “the marker junctions”, cor:marker].* (M2) Shield law. *A maximal interior run of L forced gap edges costs q^L and carries $L - 1$ isolated cycles; hence the gap-bridge kernel is $\mathfrak{g}(q, y) = q/(1 - qy)$ and the gap-marked bulk kernel is the $K_{\mathfrak{g}}$ of (4).* (M3) Decomposition. *The bivariate series of [5, §sec:bridge] is $\mathcal{W} = \mathcal{W}_+ + \mathcal{W}_- + \mathcal{W}_0$, the sectors $k^* > 0$, $k^* < 0$, $k^* = 0$, with*

$$\begin{aligned} \mathcal{W}_+ &= x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, & \mathcal{W}_- &= x^{-1} \langle \tfrac{1}{2} E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \\ \mathcal{W}_+ + \mathcal{W}_- &= \frac{1}{2x} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \end{aligned} \quad (8)$$

where $\mathcal{T} = D_e K$ is the travel operator of [5, Definition “section identity”, def:model], $D_e = \text{diag}(e_s)$ with $e_s = 2q^{1+s}$, and $\mu^{(y)}, E^{(y)}, \lambda^{(y)}$ are as in (5)–(6). The sum over the marker data $(\varepsilon^*, \delta^*)$ is already carried out in these vectors: $\delta^* = 1$ at the far site gives 2μ and $\delta^* = 0$ gives E when $k^* > 0$, and the roles are exchanged when $k^* < 0$. The $k^* = 0$ sector is $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$, with A_i built from bulk quantities alone; it contains no travel resolvent.

Proof and standing. Let $(k^*, \varepsilon^*, \delta^*, d)$ parametrise the elements of W_{univ} as in [3, Theorem “Normal form and lamp model”, thm:normal-form]. Two facts are formalised in Lean 4 with the standard axioms only:

- the affine realisation intertwines the three generators and is injective, and every element of the model is reachable, so the word length of the model is the Cayley length in W_{univ} (RJPhi.lean: `intertwine_s1, intertwine_s2, intertwine_s3, injective; reachability is reaches_of_phiZ` in PhilLipschitz.lean);
- $\text{wordLength} = \ell_R + 2c$ for every element, with ℓ_R and c the site-local closed forms `1RTrue`, `cTrue` of `CorrectedSpan.lean` (RJMetricAll.lean, theorem `metricAll`).

Given these, ℓ_R and c are sums of site and edge costs whose only non-bulk sites are 0 and k^* ([5, Corollaries `cor:localcost` and `cor:marker`]; a gap run of length L contributes $q^L y^{L-1}$ and a travel-interior site is never a cut). Summing these site costs chain by chain gives (4) with source u , the vectors (5)–(6), and the three sectors of (8); this is [5, Theorem “(M3’)”, thm:M3prime], proved in [5, Appendix `app:M3prime`], and the regrouping of $\mathcal{W}_0$ is [5, Remark “the regrouping of $\mathcal{W}_0$ ”, `M3:rem:W0regroup`]. That last step is a hand proof, with two independent adversarial reviews, and it is not formalised. Every identity holds in $\mathbb{Z}[y][[x]]$, and the series agree with enumeration as recorded in [5, §sec:bridge]. $\square$

One caveat: ℓ_R in $\mathcal{W}$ is the closed-form relaxed length of [3]. That it also equals the minimum over relaxed realisations ([3, Remark `rem:metric-formula-status`]) affects only the *interpretation* of V as a relaxed count, not U and not the series identities. How (M) was carried in earlier versions, and which of its weak links the proof removes, is recorded in [5, §sec:inputL].

$V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, q)$ ([5, §sec:bridge]), the two evaluations differing only through $\mathfrak{g}_V = \frac{q}{1-q}$, $\mathfrak{g}_U = \frac{q}{1-q^2}$ and the slot B_{0z} ; and $\lambda^{(y)}, \mu^{(y)}, \mathcal{W}_0$ are holomorphic at a travel pole q_m , by [5,

(eq:smresolvent)], as soon as $S_e(q_m) \neq 0$ and $1 - \mathfrak{g}t_1 \neq 0$ there, which is [5, Proposition prop:selower] and [5, Corollary “the gate closes”, app:gate] ($|\mathfrak{g}_V t_1| \leq 0.983$, and $|\mathfrak{g}_U t_1| < |\mathfrak{g}_V t_1|$). In particular $\mathcal{W}_0$ has no travel pole: by [5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup] it is a polynomial in B_0 with coefficients built from the bulk vector P and monomials, so its only possible singularities are at $S_e = 0$ and at $1 - \mathfrak{g}t_1 = 0$.

The growth series are defined as power series, with radius of convergence $x_1 = \sqrt{q_1} = 1/\beta_2 = 0.6704 \dots$ (Corollary 19), and the local forms below are statements about functions on a larger disc. The following lemma supplies that continuation and identifies it with the series.

Lemma 13 (meromorphic continuation). *Let y be either a complex constant with $|y| \leq 1$ or $y = x^2$. Then the specialisation $x \mapsto \mathcal{W}(x, y)$ of the series of [5, Theorem “(M3’)”, thm:M3prime] converges near $x = 0$ and is the Taylor series at 0 of a single-valued meromorphic function on the disc $|x| < 1$, given there by (8) and $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$ read as analytic expressions. Its poles in $0 < |x| < 1$ lie over points $q = x^2$ at which $1 - \Sigma_1(q) = 0$, $S_e(q) = 0$ or $1 - \mathfrak{g}(q, y) t_1(q) = 0$. In particular $V = \mathcal{W}(x, 1)$ and $U = \mathcal{W}(x, x^2)$ are meromorphic on $|x| < 1$, and $G_0^T = \Sigma_0/(1 - \Sigma_1)$ is meromorphic on $|q| < 1$. (The continuation concerns the explicit operators of [5, §§sec:sect–sec:mob]; its identification with U and V is Theorem 12.)*

Proof. Put $q = x^2$, so $|q| < 1$; $\mathfrak{g} = q/(1 - qy)$ is holomorphic on the disc, since $|qy| < 1$ in both cases.

Operators. $q \mapsto \mathcal{T}(q)$ and $q \mapsto M_0(q)$ are holomorphic on $|q| < 1$ as maps into the bounded operators on ℓ^1 , in operator norm: the row truncations M_N in the proof of [5, Lemma “compactness and meromorphy”, lem:fred] have finitely many non-zero rows, each a norm-convergent ℓ^∞ -valued power series in q (the row s is $e_s \sum_{s'} q^{\max(s, s')} \delta_{s'}$), hence are holomorphic in norm, and, with M either operator and $\vartheta \in \{0, 1\}$ its exponent offset in [5, Definition “section identity”, def:model], $\|M - M_N\| \leq \sum_{s > N} 2|q|^{\vartheta+s} \rightarrow 0$ uniformly on $|q| \leq r < 1$; a locally uniform norm limit of holomorphic operator-valued maps is holomorphic. With compactness ([5, Lemma “compactness and meromorphy”, lem:fred]) and invertibility near $q = 0$ ([5, Lemma “existence, uniqueness, analyticity”, lem:modelexist]), the analytic Fredholm theorem makes $(I - \mathcal{T})^{-1}$ and $R_0 = (I - M_0)^{-1}$ meromorphic on $|q| < 1$ in operator norm, with poles only where $1 - \Sigma_1 = 0$, resp. $S_e = 0$ ([5, Corollary “singularities of the two gapless resolvents”, cor:sing]).

Bulk quantities. $u = (2q^{2b})_b$ is a norm-convergent, hence holomorphic, ℓ^1 -valued function, so $\Psi = R_0 u$ is meromorphic ℓ^1 -valued and $t_1 = v^\top \Psi$ is meromorphic. $1 - \mathfrak{g}t_1$ tends to 1 as $x \rightarrow 0$, so it is not identically zero, and $B_0 = 1/(1 - \mathfrak{g}t_1)$, $P = B_0 \Psi$, $\mathfrak{g}S = B_0 - 1$ and $B_{0z} = 1 + y(B_0 - 1)$ are meromorphic ([5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup]).

Boundary vectors and the sectors $k^ \neq 0$.* Every entry of [5, (eq:app-mu)–(eq:app-lambda)] is a combination of B_0 , B_{0z} and sums $\sum_b P_b x^{m_b}$ with $m_b \geq 0$, so $|\mu_s^{(y)}|, |E_s^{(y)}|, |\lambda_s^{(y)}| \leq C(|B_0| + |B_{0z}| + \|P\|_{\ell^1})$ uniformly in s , locally uniformly in x away from the poles of B_0 and Ψ . Hence $D_e \lambda^{(y)}$ has ℓ^1 -tails $\sum_{s > N} |e_s \lambda_s^{(y)}| = O(|q|^{N+2})$, locally uniformly, and is meromorphic ℓ^1 -valued; so is $(I - \mathcal{T})^{-1} D_e \lambda^{(y)}$. Its pairing with the bounded vectors $\mu^{(y)}$ and $\frac{1}{2} E^{(y)}$ is an absolutely convergent sum, locally uniformly convergent away from the poles (an ℓ^1 -valued continuous map has uniformly small tails on compact sets), hence meromorphic. This gives $\mathcal{W}_\pm$; the prefactor x^{-1} is at most a pole at $x = 0$, and is removed below.

The sector $k^ = 0$.* By [5, Remark “the regrouping of $\mathcal{W}_0$ ”, M3:rem:W0regroup], $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$. The A_i are not finite sums: each is an absolutely convergent double sum, over the left and the right chain kinds $\mathbf{E}, \mathbf{Z}, \mathbf{N}_{\pm, b}$ of [5, (eq:app-W0)] ($b \geq 1$), of monomials $x^j y^\chi$ with $j \geq 0$, $\chi \in \{0, 1\}$, times at most two entries of Ψ . Since $|x^j y^\chi| \leq 1$, the double sum is dominated by a constant times $(1 + \|\Psi\|_{\ell^1})^2$, converges locally uniformly where Ψ is holomorphic, and is meromorphic.

Identification with the x -adic series. Choose $r_0 > 0$ so small that on $|x| < r_0$ the operators $\mathcal{T}$, M_0 and $M_0 + \mathbf{g}uv^\top$ have ℓ^1 -norm below $\frac{1}{2}$ ([5, Lemma “existence, uniqueness, analyticity”, lem:modeexist]); the rank-one term has norm at most $|\mathbf{g}| \|u\|_{\ell^1}$ and $|\mathbf{g}t_1| < \frac{1}{2}$. There every object above is the sum of a norm-convergent Neumann series $((I - \mathcal{T})^{-1}, R_0, P = (I - M_0 - \mathbf{g}uv^\top)^{-1}u$, and $B_0 = \sum_n (\mathbf{g}t_1)^n$), whose terms are holomorphic on $|x| < r_0$ and whose Taylor series at 0 are the terms of the x -adic Neumann series of [5, Lemma M3:lem:conv], of x -adic order tending to infinity. If $\sum_n h_n$ converges uniformly on a disc about 0, the Taylor coefficients of the sum are the sums of the Taylor coefficients of the h_n (Weierstrass); when $\text{ord } h_n \rightarrow \infty$ each of these sums is finite and is the x -adic sum. Applying this to each Neumann series, to the absolutely convergent pairings and to the double sums A_i , the Taylor series at 0 of the analytic $\mathcal{W}(x, y)$ is the x -adic series of [5, Theorem “(M3’)”, thm:M3prime], specialised at y (the specialisation $\mathbb{Z}[y][[x]] \rightarrow \mathbb{C}[[x]]$ at a constant y , or at $y = x^2$, is x -adically continuous and commutes with the sums). In particular that function is holomorphic at $x = 0$. At $y = 1$ the series is V , and at $y = x^2$ it is U ([5, Corollary cor:M3prime-growth]). A power series that agrees near 0 with a function meromorphic on the disc is continued by it, uniquely. For G_0^T , Σ_0 and $1 - \Sigma_1 \neq 0$ are holomorphic on $|q| < 1$ ([5, Theorem thm:blocks], first clause). $\square$

Proposition 14 (the shape of the local form). *Let q_m be a travel pole, let $p \geq 1$ be the order of the zero of $1 - \Sigma_1$ there (finite, Lemma 7), and let π be the order of the pole of $(I - \mathcal{T})^{-1}$ there (finite, [5, Lemma “compactness and meromorphy”, lem:fred]). Then $\pi \geq p \geq 1$ and, for each y ,*

$$\mathcal{W}(x, y) = \frac{\mathcal{N}_y}{(q - q_m)^\pi} + (\text{poles of order } < \pi) + H_m, \quad H_m \text{ holomorphic at } q_m,$$

with the leading numerator a perfect square

$$\mathcal{N}_y = \frac{c_\star}{2x_m} \Pi_y(q_m)^2, \quad \Pi_y(q_m) := \langle \lambda^{(y)}, R \rangle, \quad c_\star \neq 0, \quad x_m = \sqrt{q_m}, \quad (9)$$

where R spans $\ker(I - \mathcal{T}(q_m))$, normalised by its sections,

$$\sum_{s \geq 0} q^{ks} R_s = \Sigma_k(q_m) \quad (k \geq 0), \quad \text{in particular } \sum_s R_s = \Sigma_0(q_m), \quad \sum_s q^s R_s = \Sigma_1(q_m) = 1,$$

and L spans $\ker(I - \mathcal{T}(q_m)^\top)$, explicitly $L_s = R_s/e_s$. There are no cross terms between the sectors. In particular $\Pi_y(q_m) \neq 0$ implies that q_m is a pole of $\mathcal{W}(\cdot, y)$, of order exactly π .

Proof. Write $\mathcal{R}(q) = (I - \mathcal{T}(q))^{-1} = \sum_{j=1}^\pi A_j(q - q_m)^{-j} + \text{hol}$, with $A_\pi \neq 0$. Multiplying $(I - \mathcal{T}(q))\mathcal{R}(q) = I$ by $(q - q_m)^\pi$ and letting $q \rightarrow q_m$ gives $(I - \mathcal{T}(q_m))A_\pi = 0$, so $\text{ran } A_\pi \subseteq \ker(I - \mathcal{T}(q_m))$, which is one-dimensional by [5, Lemma “the null space is at most one-dimensional”, lem:nullspace]; the same step on $\mathcal{R}(q)(I - \mathcal{T}(q)) = I$ gives $A_\pi(I - \mathcal{T}(q_m)) = 0$. Hence $A_\pi x = c_\star \langle L, x \rangle R$ with $c_\star \neq 0$, R spanning the kernel and L the cokernel. The normalisation of R is [5, Lemma “the null space is at most one-dimensional”, lem:nullspace]: with the sections $G_k^R = \sum_s q^{ks} R_s$, one has $G_k^R = G_1^R \Sigma_k$ and $\Sigma_1(q_m) = 1$, so $G_1^R = 1$ gives $G_k^R = \Sigma_k(q_m)$. That $L_s = R_s/e_s$ spans the cokernel follows from $\mathcal{T} = D_e K$ with $D_e = \text{diag}(e_s)$ and K symmetric: from $D_e K R = R$ one gets $K D_e (D_e^{-1} R) = K R = D_e^{-1} R$, that is $\mathcal{T}^\top L = L$, and $L \in \ell^\infty$ because $|R_s| \leq |e_s| \|R\|_{\ell^1}$; the cokernel is one-dimensional by Fredholm. Substituting the Laurent expansion into $\mathcal{W}_+ + \mathcal{W}_-$ of (8), and using that $\lambda^{(y)}, \mathcal{W}_0$ are holomorphic at q_m , the leading coefficient is $(2x_m)^{-1} \langle \lambda, A_\pi D_e \lambda \rangle = (2x_m)^{-1} c_\star \langle L, D_e \lambda \rangle \langle \lambda, R \rangle = (2x_m)^{-1} c_\star \langle \lambda, R \rangle^2$, since $D_e L = R$. It is non-zero exactly when $\Pi_y(q_m) \neq 0$, and then q_m is a pole of order π . Finally $\langle \mathbf{1}, \mathcal{R}E \rangle = \Sigma_0/(1 - \Sigma_1)$ ([5, Proposition “the travel k -sum telescopes”, prop:travelsum]) has a pole of order exactly p at q_m , since $\Sigma_0(q_m) \neq 0$ (Proposition 11), while a matrix element of $\mathcal{R}$ has pole order at most π ; hence $\pi \geq p$. $\square$

4.3 The junction pairing (R-J)

(R-J) is the statement that $\Pi_y(q_m) = \langle \lambda^{(y)}, R \rangle \neq 0$ at infinitely many travel poles, for $y = 1$ and $y = q$. Earlier versions carried it as a hypothesis. It is now a theorem.

Theorem 15 (closed form of the junction pairing). *Let q_m be a travel pole, $x = \sqrt{q_m}$, and R normalised as in Proposition 14. Then*

$$(1 - \mathfrak{g}_V t_1) \Pi_1(q_m) = \frac{2q(1+x)}{1-q} \left(t_1 + \frac{1+x}{2x} \right), \quad (10)$$

$$\Pi_q(q_m) = \frac{2q B_0(1+x)}{1-x} \left(t_1 \frac{1-x+q}{1+q} + \frac{1}{2x} \right), \quad B_0 = \frac{1}{1 - \mathfrak{g}_U t_1}. \quad (11)$$

Consequently $\Pi_1(q_m) > 0$ and $\Pi_q(q_m) > 0$ at every travel pole q_m .

Proof. Throughout $q = q_m$.

Reduction to the ungapped bulk vector. Put $\Psi_0 = 1$ and let $\Psi = (\Psi_b)_{b \geq 1} = R_0 u$ be the solution of the bulk system with $\mathfrak{g} = 0$ and source u , that is

$$\Psi_b = 2q^b F_b, \quad F_t := \sum_{b \geq 0} q^{\max(b,t)} \Psi_b \quad (b \geq 1),$$

the term $b = 0$ of F_t being the source q^t . Then $t_1 = v^\top R_0 u = \sum_{b \geq 1} q^b \Psi_b$ and, by [5, (eq:smresolvent)], $P = R_0 u / (1 - \mathfrak{g} t_1) = B_0 \Psi$, where $B_0 = 1 / (1 - \mathfrak{g} t_1)$ ([5, Remark “the bulk sum of the model”, rem:bulksource]). So every bulk weight in (5)–(6) is $B_0 \Psi_b$, with $b = 0$ standing for the empty and gap-opening chains. In the variable $q = x^2$,

$$\begin{aligned} x^{\max(2s,2b)} &= q^{\max(s,b)}, & x^{\max(2s+2,2b)} &= q q^{\max(s,b-1)}, \\ x^{\max(2s+1,2b-1)} &= x q^{\max(s,b-1)}, & x^{\max(2s+1,2b+1)} &= x q^{\max(s,b)}, \end{aligned}$$

with $\max(s, -1) = s$. At $y = 1$ the slot B_{0z} equals B_0 , and (5), (6) combine into

$$\lambda_s^{(1)} = B_0 \sum_{b \geq 0} \Psi_b \left((1+x) q^{\max(s,b)} + (q+x) q^{\max(s,b-1)} \right), \quad (12)$$

hence $\langle \lambda^{(1)}, R \rangle = B_0((1+x)X + (q+x)Y)$ with

$$X = \sum_{b \geq 0} \Psi_b (KR)_b, \quad Y = \sum_{b \geq 0} \Psi_b G_b, \quad (KR)_b = \sum_{t \geq 0} q^{\max(b,t)} R_t, \quad G_b = \sum_{s \geq 0} q^{\max(s,b-1)} R_s.$$

All double sums converge absolutely ($\Psi \in \ell^1$ with $|\Psi_b| = O(q^{2b} |\Psi|_{\ell^1})$, $R \in \ell^1$), so they may be summed in either order.

Two facts about R . From $\mathcal{T}R = R$, $R_s = 2q^{s+1}(KR)_s$ for every $s \geq 0$; and the normalisation $\sum_s q^s R_s = \Sigma_1(q_m) = 1$ of Proposition 14 says $(KR)_0 = G_0 = 1$. In particular $R_0 = 2q$.

The X shift identity. Summing first over t and splitting off $b = 0$, $X = 1 + S$ with $S = \sum_{b \geq 1} \Psi_b (KR)_b$. Summing first over b instead, $X = \sum_t R_t F_t$. The term $t = 0$ is $R_0 F_0 = 2q(1+t_1)$, since $F_0 = \sum_b q^b \Psi_b = 1 + t_1$; for $t \geq 1$, $R_t F_t = 2q^{t+1}(KR)_t \cdot \Psi_t / (2q^t) = q \Psi_t (KR)_t$. Hence $X = 2q(1+t_1) + qS$, and eliminating S ,

$$X = 1 + \frac{X - (1+t_1)R_0}{q}, \quad \text{so} \quad X = \frac{q(1+2t_1)}{1-q}.$$

The Y shift identity. Splitting off $b = 0$ and using $G_b = (KR)_{b-1} = R_{b-1}/(2q^b)$ for $b \geq 1$, $Y = 1 + \sum_{s \geq 0} \Psi_{s+1} R_s / (2q^{s+1})$. Summing first over b instead, and using $q^{\max(s, b-1)} = q^{-1} q^{\max(s+1, b)}$, $Y = \sum_s R_s F_{s+1} / q = \sum_s R_s \Psi_{s+1} / (2q^{s+2})$. The two expressions give

$$Y = 1 + qY, \quad \text{so} \quad Y = \frac{1}{1-q}.$$

The closed forms. With $q = x^2$, $(1+x)X + (q+x)Y = ((1+x)q(1+2t_1) + q+x)/(1-q)$, and since $(1+x)q + q + x = x(1+x)^2$ this is (10). At $y = q$ only the slot $s = 0$ of $E^{(q)}$ changes, from B_0 to $B_{0z} = 1 + q \mathbf{g}_U S = B_0(1 - t_1 q / (1+q))$ (using $\mathbf{g}_U S = B_0 - 1 = \mathbf{g}_U t_1 B_0$). So $\Pi_q = B_0((1+x)X + (q+x)Y) + (B_{0z} - B_0)R_0$, now with $B_0 = 1/(1 - \mathbf{g}_U t_1)$, and collecting terms with $(1+x)^2(1-x+q) = (1+x)(1+q) - q(1-q)$ (both equal $1+x+x^3+x^4$) gives (11).

Positivity for $\tau_m \leq 5 \cdot 10^{-3}$. By [5, Corollary “the gate closes”, app:gate], $|\mathbf{g}_V t_1| \leq 0.983$, and $\mathbf{g}_U < \mathbf{g}_V$. Hence $1 - \mathbf{g}_U t_1 \geq 0.017$ for both y , and $q|t_1| \leq 0.983(1-q)$, so $|t_1| < 0.12$ once $q \geq 0.9$ (here $q \geq e^{-0.005}$). Since $(1+x)/(2x) \geq 1$, $1/(2x) > 1/2$ and $0 < (1-x+q)/(1+q) < 1$, both brackets are positive, and so are the prefactors.

Positivity at the poles above that threshold. This part is a computer-assisted certificate: at $q_1, \dots, q_{12}$, Π_1 and Π_q are enclosed in MPFR interval arithmetic and found positive, together with $1 - \mathbf{g}_U t_1 > 0$ (code/zeta_probe/rj_certificates/r35rust/src/bin/r54b.rs). The pole index comes from code/zeta_probe/rj_certificates/r39pole, an interval-arithmetic zero count of $1 - \Sigma_1$ which isolates exactly 13 zeros on $[0, 0.9988]$ (the right endpoint taken as its binary64 value), each in its own cell. Its tiling of the interval was repaired after a first review found gaps of order 10^{-77} between cells, and the repaired code was rebuilt and re-run from current source by a second reviewer; the trusted base is MPFR and the tool’s own code. $\square$

Only infinitely many poles are needed for Theorems 18 and 20, and the analytic part ($\tau_m \leq 5 \cdot 10^{-3}$) supplies them; the certificate is needed only for the statement “at every travel pole”. The algebraic core is formalised in Lean 4 ([5, §sec:RJlean]); the eigen-relations of R , the gate and the analytic estimates behind it enter the formal statements as hypotheses and are not formalised.

Corollary 16 ((R-J) at the dominant pole). $\Pi_1(q_1) > 0$ and $\Pi_q(q_1) > 0$; this is the case $m = 1$ of Theorem 15.

5 Transcendence of the travel resolvent, of V and of U

The relaxed series is the travel resolvent dressed by the bulk. The resolvent itself is settled unconditionally; passing from it to V uses the transfer model (Theorem 12) and the junction pairing (Theorem 15).

Theorem 17 (the travel resolvent). *The travel resolvent $G_0^T = \Sigma_0 / (1 - \Sigma_1)$ is transcendental over $\mathbb{Q}(x)$. This is unconditional: it uses no growth hypothesis, no asymptotic theorem, and no statement about the strand-walk model. With the pole set taken from Proposition 6, as in the proof below, it has no numerical input. The second route to the pole set, [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles], consumes only $|T_2(m\pi)| < 1$ from the contour bound of [5, Lemma “absolute bound on T_2 ”, lem:T2abs], against a computed worst value 0.027; that bound is a verified enclosure over the parameter region, not a sampled one.*

Proof. Σ_0 and $1 - \Sigma_1$ are holomorphic on $|q| < 1$ ([5, Theorem thm:blocks], first clause) and $1 - \Sigma_1 \not\equiv 0$. The zero set of $1 - \Sigma_1$ in $(0, 1)$ is infinite and accumulates exactly at $q = 1$ (Proposition 6(c) together with Lemma 7; alternatively [5, Lemma “infinitude and accumulation of the travel poles”,

lem:infpoles], which consumes from [5, Lemma “absolute bound on T_2 ”, lem:T2abs] only $|T_2| < 1$). At each such q_m the numerator does not vanish, by Proposition 11. Hence every q_m is a pole of G_0^T , of order the vanishing order of $1 - \Sigma_1$ there, which is finite. So G_0^T has infinitely many poles in $|q| < 1$ accumulating at $|q| = 1$; under $q = x^2$ these are infinitely many poles in $|x| < 1$ accumulating at $|x| = 1$. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$, so no algebraic function has such a pole set; hence G_0^T is transcendental. It is not D -finite either, in q or in x , and $|q| = 1$ is its natural boundary (Theorems 26 and 30): for D -finiteness the finiteness of the pole set of an algebraic function is replaced by Lemma 25, which confines the poles in the disc of a solution of a linear differential equation to the finitely many zeros of its leading coefficient. $\square$

Theorem 18. *The relaxed growth series V is transcendental over $\mathbb{Q}(x)$. (Standing: Theorems 12 and 15.)*

Proof. By Lemma 13, V is meromorphic on $|x| < 1$, and the local forms below are those of this continuation. By Proposition 14, at a travel pole q_m the relaxed series has the local form $V = \mathcal{N}_1(q - q_m)^{-\pi} + (\text{poles of order } < \pi) + H_m$ with H_m holomorphic, $\pi \geq 1$, and $\mathcal{N}_1 = (2x_m)^{-1} c_\star \Pi_1(q_m)^2$ with $c_\star \neq 0$. That proposition consumes two non-vanishing statements, and they are separate: $S_e(q_m) \neq 0$, in the quantitative form $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ of [5, Proposition prop:selower], which makes the bulk boundary vectors $\lambda^{(1)}, \mu^{(1)}$ holomorphic at q_m and says nothing about vanishing; and $\Sigma_0(q_m) \neq 0$, from Proposition 11, which pins the pole order $\pi \geq p \geq 1$; and $\Pi_1(q_m) \neq 0$ at every travel pole, by Theorem 15.

So the leading Laurent coefficient is non-zero at every travel pole, and V has a pole at each of them. The travel poles are infinite in number and accumulate at $q = 1$, that is at $|x| = 1$ (Proposition 6 and Lemma 7); this is the boundary of the disc $|x| < 1$ of meromorphy, not of the disc of convergence $|x| < x_1$, which only the dominant pole bounds. They are locally finite in $(0, 1)$ (Lemma 7), so an infinite subset of them accumulates at $q = 1$ and nowhere else. Hence V has infinitely many poles in $|q| < 1$ accumulating at $|q| = 1$; under $q = x^2$ these are infinitely many poles in $|x| < 1$ accumulating at $|x| = 1$. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$, so an infinite pole set accumulating at the boundary is impossible for one; hence V is transcendental. $\square$

Corollary 19. *The exponential growth rate of v_n is $\beta_2 = 1.4916177871\dots$, with β_2^2 the reciprocal of the smallest travel pole q_1 ; the value $\frac{3}{2}$ is excluded. It needs only the location of the dominant travel pole: not [5, Remark “the sharp size of T_2 ; HEURISTIC”, lem:cos], not [5, Appendix app:annulus], and not [5, Appendix app:star]. It does share with Theorem 18 the transfer model (M), which is what places q_1 on V rather than on G_0^T alone, and at the dominant pole the junction pairing is positive (Corollary 16). The lower bound β_2 for u_n and v_n is free even of (M), coming from the pure-travel subcount [3]. The same rate for u_n follows from Proposition 1 (the former hypothesis (T) of Theorem 20, now proved) together with the positivity of $\Pi_q(q_1)$; $u_n \leq v_n$ gives the upper bound unconditionally.*

Whether the number β_2 is transcendental is a separate question, open, and with a clean reduction to a value statement for a Hahn–Exton q -Bessel function; it is treated in [5, Appendix app:beta2], which is independent of everything else in this paper.

5.1 The true series U

Theorem 20. *The growth series U of W_{univ} is transcendental over $\mathbb{Q}(x)$. (Standing: Theorems 12 and 15.)*

Remark 21 (the inputs of Theorem 20). The travel poles pass from V to U by Proposition 1, whose proof does not use the metric identity. The factorisation [5, (eq:mobius)] with the dictionary [5, (eq:dict)] and the reciprocity $t_0 = b_1$ is [5, Theorem “the bulk dictionary at the travel poles”, thm:L]. The residue is the square of $\Pi_q(q_m)$, which is Theorem 15. The history of the former inputs (T) and (L) is recorded in [5, §§sec:inputT–sec:inputL].

Independently of (M), (T) and (R-J), the half $|s| < 1$ of the gate (7) holds at every travel pole. The numerator amplitude estimate $(\star)$ (at every travel pole q_m , $|P_{12}(q_m)| w_m^3 < 2$) is [5, Theorem “the amplitude estimate $(\star)$ ”, thm:star] ([5, Appendix app:star]): $|P_{12}| \leq 0.619 \tau^{3/2}$ for $\tau \leq 5 \cdot 10^{-3}$ and $|P_{12}| w^3 \leq 0.75$ at the six poles above that threshold. The passage to $|s| < 1$ is [5, Corollary “the gate closes”, app:gate]: $|s| \leq 0.983$ on the uniform range and $|s| \leq 0.296$ at the six exceptional poles, with the denominator lower bound $|S_e| \geq 0.63\sqrt{\tau}$ supplied by [5, Lemma “the denominator at the pole”, app:Se]. The remaining half $b_0 > 0$ is proved in the same corollary from a sign chain through the closed form for b_0 , which is [5, Corollary “the dictionary”, cor:dict]. The leading-order picture of [5, Remark “derivation of the U -numerator amplitude”, rem:elemY3] (the phase-matched zero giving $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$, a fourfold margin below the threshold $1/\sqrt{2}$) is consistent with, and explained by, the second-difference identity of [5, Proposition prop:P12second].

Proof. By Lemma 13 at $y = x^2$, U is meromorphic on $|x| < 1$, and the local forms below are those of this continuation. By Proposition 1 (the former input (T)), U inherits the travel poles. By Proposition 14 at $y = q$, at each travel pole U has the local form $\mathcal{N}_q(q - q_m)^{-\pi} + (\text{lower order}) + H_m$ with $\mathcal{N}_q = (2x_m)^{-1} c_\star \Pi_q(q_m)^2$ and $c_\star \neq 0$; the boundary vectors are holomorphic there because the gate (7) holds at *every* travel pole. Indeed, on the uniform range $\tau \leq 5 \cdot 10^{-3}$, $|P_{12}| \leq 0.619 \tau^{3/2}$ ([5, Theorem “the amplitude estimate $(\star)$ ”, thm:star]) and $|S_e| \geq 0.63\sqrt{\tau}$ ([5, Lemma “the denominator at the pole”, app:Se]) give $|s| = \frac{q}{1-q} |P_{12}|/|S_e| \leq \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1$, and $b_0 > 0$ follows from the sign chain of the same lemmas through the closed form of [5, Corollary “the dictionary”, cor:dict]; the six exceptional poles are checked directly ([5, Corollary “the gate closes”, app:gate]). By Theorem 15 the leading coefficient $\mathcal{N}_q$ is non-zero at every q_m , so U has an uncanceled pole at each of them; these accumulate at $q = 1$ (Proposition 6 and Lemma 7), so U has infinitely many poles in the unit disc accumulating at its boundary and is transcendental over $\mathbb{Q}(x)$, as in Theorem 18. Transcendence does not consume the exceptional-pole enumeration of [5, Theorem “the amplitude estimate $(\star)$ ”, thm:star]: the uniform range $\tau \leq 5 \cdot 10^{-3}$ alone carries infinitely many poles (by Proposition 6 and Lemma 7, only finitely many travel poles lie outside it); the enumeration only upgrades “at infinitely many travel poles” to “at every travel pole” in the gate. For orientation: the computed values over $m = 5, \dots, 71$ are $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.1768$ and $s \rightarrow \frac{1}{4}$, well inside every bound above. $\square$

Remark 22 (the standing of Theorem 20). The unconditional content of this section is the gate: at every travel pole, $|P_{12}| \leq 0.619 \tau^{3/2}$ ([5, Theorem “the amplitude estimate $(\star)$ ”, thm:star]), $|S_e| \geq 0.63\sqrt{\tau}$ ([5, Lemma “the denominator at the pole”, app:Se]) and hence $|s| \leq 0.983 < 1$ ([5, Corollary “the gate closes”, app:gate]), together with the infinitude and accumulation of the travel poles (Proposition 6, Lemma 7) and the fact that these facts compose to non-algebraicity ([5, Remark “the assembly is machine-checked”, rem:uasembly]). The factorisation [5, (eq:mobius)], the reciprocity $t_0 = b_1$, the dictionary [5, (eq:dict)] and the half $b_0 > 0$ of the gate are [5, Theorem “the bulk dictionary at the travel poles”, thm:L]; the junction pairing is Theorem 15.

(T) is now proved. Proposition 1 follows from [5, Corollary cor:localzero]: a pure-travel element has no gap edge, so by [5, Theorem “no gap edge forces $c = 0$ ”, thm:nogap] some relaxed-optimal realisation is a single open walk, which bounds $\ell_T \leq \ell_R$; the reverse inequality is immediate. The metric identity $\ell_T = \ell_R + 2c$ is not used.

(R-J) is Theorem 15, and (M) is Theorem 12. The standing of Theorem 20 is therefore that of its inputs: the gate ([5, Appendix app:star], which contains verified interval-arithmetic enclosures); the metric identity and the faithfulness of the model (Lean); the assembly (M3) (a hand proof, [5, Appendix app:M3prime], with two independent adversarial reviews and exact series agreement to x^{26}); and, for the statement at every travel pole rather than infinitely many, the interval certificate at the first twelve poles. The assembly is not formalised.

6 Beyond algebraicity: D -finiteness, functional equations and the natural boundary

Theorems 18 and 20 exclude algebraic equations. The pole set that proves them excludes much more, because it is infinite and lies inside the disc $|x| < 1$ on which U and V are meromorphic (Lemma 13), well beyond their radius of convergence $x_1 = 0.6704\dots$. This section records what it excludes. Every statement below about U or V has the standing of Theorems 18 and 20 (Theorems 12 and 15); every statement about the travel resolvent G_0^T alone is unconditional, as Theorem 17 is.

6.1 Poles of U and V on the negative axis

Proposition 23. *At every travel pole q_m , U has a pole at $x = -x_m$ as well as at $x = x_m$, both of order π , the order of the pole of $(I - \mathcal{T})^{-1}$ at q_m . At $x = -x_m$ the junction pairing is negative:*

$$\Pi_q|_{x=-x_m} = \frac{2q_mB_0(1-x_m)}{1+x_m} \left(t_1 \frac{1+x_m+q_m}{1+q_m} - \frac{1}{2x_m} \right) < 0, \quad B_0 = \frac{1}{1-\mathfrak{g}_U t_1}.$$

(Standing: Theorems 12 and 15; at the six poles with $\tau_m > 5 \cdot 10^{-3}$, the enumeration standing of [5, Theorem “the amplitude estimate ($\star$)”, thm:star].)

Proof. The travel resolvent $(I - \mathcal{T})^{-1}$, the bulk vector Ψ , t_1 and B_0 depend on x only through $q = x^2$; x itself enters $\mathcal{W}(x, q)$ only through the monomials of [5, (eq:app-mu)–(eq:app-E)] and the prefactor x^{-1} . The proofs of Proposition 14 and of (11) use only $q = x^2$, never the sign of x (the reductions $x^{\max(2s+1, 2b+1)} = x q^{\max(\cdot, \cdot)}$ and the final polynomial identities hold for either root). Near $x = -x_m$ one has $q - q_m = (x + x_m)(x - x_m)$ with the second factor non-zero, so $U = \mathcal{N}_q(-2x_m)^{-\pi}(x + x_m)^{-\pi} + (\text{lower order})$ with $\mathcal{N}_q = c_\star(-2x_m)^{-1}\Pi_q|_{x=-x_m}^2$, and (11) at $x = -x_m$ is the displayed formula. The prefactor is positive: $0 < x_m < 1$, and $1 - \mathfrak{g}_U t_1 > 0$ because $|\mathfrak{g}_U t_1| < |\mathfrak{g}_V t_1| = |s| < 1$ ([5, Corollary “the gate closes”, app:gate]). For the bracket write $t_1 = s(1 - q)/q$ and use $(1 + x + q)/(1 + q) = 1 + x/(1 + x^2) \leq \frac{3}{2}$:

$$2x_m |t_1| \frac{1 + x_m + q_m}{1 + q_m} \leq 3|s| \frac{1 - q_m}{x_m}.$$

For $\tau_m \leq 5 \cdot 10^{-3}$, $|s| \leq 0.983$ and $(1 - q_m)/x_m \leq \tau_m e^{\tau_m/2} \leq 0.00502$, so the right side is below 0.015. At the six poles with $\tau_m > 5 \cdot 10^{-3}$, $|s| \leq 0.296$ ([5, Corollary “the gate closes”, app:gate]) and $(1 - q_m)/x_m \leq (1 - q_1)/x_1 = 0.8212$, since $q \mapsto (1 - q)/\sqrt{q}$ decreases; the right side is at most 0.730. In both cases $|t_1|(1 + x_m + q_m)/(1 + q_m) < 1/(2x_m)$, the bracket is negative, $\mathcal{N}_q \neq 0$, and $-x_m$ is a pole of order exactly π . $\square$

At the dominant pole the bracket is -0.2155 , and it tends to $-\frac{1}{2}$ along the ladder. Only the uniform range is needed for infinitely many poles at $-x_m$; the six-pole list upgrades this to every travel pole.

Proposition 24 (V at $-x_m$). *At every travel pole q_m , V has a pole at $x = -x_m$ of order π . With $\beta_m := t_1 - (1 - x_m)/(2x_m)$,*

$$(1 - \mathbf{g}_V t_1) \Pi_1|_{x=-x_m} = \frac{2q_m(1 - x_m)}{1 - q_m} \beta_m,$$

and for $\tau = \tau_m \leq 5 \cdot 10^{-3}$, with $h = \tau/2$, $\gamma = 4 \sinh h + 2 - 2 \cosh h$ and $\Lambda = e^{h/2} \sinh h / (2h(4 - \gamma/3))$,

$$\frac{1 - 5.14\sqrt{\tau}}{1 + 1.20\sqrt{\tau}} \leq \frac{\beta_m}{\Lambda\tau^2} \leq \frac{1 + 5.14\sqrt{\tau}}{1 - 1.20\sqrt{\tau}}, \quad \text{so} \quad 0.073\tau^2 \leq \beta_m \leq 0.187\tau^2.$$

At the six poles with $\tau_m > 5 \cdot 10^{-3}$, direct evaluation gives $\beta_m = 0.1168, 1.119 \cdot 10^{-3}, 1.362 \cdot 10^{-4}, 3.484 \cdot 10^{-5}, 1.266 \cdot 10^{-5}, 5.651 \cdot 10^{-6}$. (Standing: Theorems 12 and 15; at the six poles with $\tau_m > 5 \cdot 10^{-3}$, the enumeration standing of [5, Theorem “the amplitude estimate ($\star$)”, thm:star].)

Here the gate does not decide the sign, unlike for U : $t_1 = s(e^\tau - 1)$ with $s \rightarrow \frac{1}{4}$, and $(1 - x_m)/(2x_m) = (e^{\tau/2} - 1)/2 = \frac{\tau}{4} + O(\tau^2)$, so the two terms of β_m cancel at leading order. The proof computes the $O(\tau^2)$ remainder. It uses an exact second-order q -difference equation for the q -cosine to eliminate the unknown curvature at the zero.

Proof. The pole. $V = \mathcal{W}(x, 1)$, and the proof of Proposition 23 applies verbatim at $y = 1$: the local form of Proposition 14 and (10) use only $q = x^2$. Near $x = -x_m$ the leading coefficient is $c_\star(-2x_m)^{-1} \Pi_1|_{x=-x_m}^2$ with $c_\star \neq 0$, and (10) at $x = -x_m$ is the displayed identity. Its prefactor is positive, and $1 - \mathbf{g}_V t_1 = 1 - s > 0$ by the gate ([5, Corollary “the gate closes”, app:gate]). So $\Pi_1|_{x=-x_m} \neq 0$ as soon as $\beta_m \neq 0$, and then $-x_m$ is a pole of order exactly π . At the six exceptional poles, $\beta_m > 0$ is the listed evaluation. It remains to bound β_m for $\tau \leq 5 \cdot 10^{-3}$.

Notation. Let $c_j = (-1)^j q^{j^2+j} / (q; q)_{2j}$ and $C(y) = \sum_j c_j y^j$, and write $\psi(t) = \cos(Ze^t; q^2) = C(Z^2 e^{2t})$. Also write $\varphi(t) = e^{-t/2} \psi(t)$, $w = \sqrt{2/\tau}$, $A = \varphi'(0) = \psi'(0)$ and $M_k = \max_{|t| \leq \tau} |\varphi^{(k)}(t)|$. At the pole, $\psi(0) = 0$ (Proposition 2), so $\varphi(0) = 0$. Moreover $z_0 = Ze^{-h}$, $qZ = Ze^{-2h}$ and $x_m = e^{-h}$.

Step 1: β_m as a symmetric sum. We have $t_1 = P_{12}/S_e$ (§4.1), $S_e = \psi(-h)$ (Proposition 2) and $P_{12} = -\frac{1}{2}[\psi(h) + \psi(-h)]$ ([5, Proposition prop:P12second]). With $(1 - x_m)/(2x_m) = (e^h - 1)/2$ this gives

$$\beta_m = -\frac{\psi(h) + e^h \psi(-h)}{2\psi(-h)} = -\frac{e^{h/2} E_1}{2S_e}, \quad E_k := \varphi(kh) + \varphi(-kh). \quad (13)$$

Step 2: a q -difference equation. The coefficients satisfy $(1 - q^{2j-1})(1 - q^{2j})c_j = -q^{2j}c_{j-1}$. Summed against y^j , this gives $C(y) - (1 + q^{-1})C(q^2 y) + q^{-1}C(q^4 y) = -q^2 y C(q^2 y)$. In the variable t , $y \mapsto q^2 y$ is $t \mapsto t - 2h$. We shift by $2h$ and use $Z^2 = 2(e^{2h} - 1)$. Then we divide by $e^{t/2+h}$, which gives, for all real t ,

$$\varphi(t + 2h) + \varphi(t - 2h) = (2 \cosh h - 4 \sinh h e^{2t}) \varphi(t). \quad (14)$$

At $t = 0$, and after differentiating twice at $t = 0$, since $\varphi(0) = 0$:

$$E_2 = 0, \quad \varphi''(2h) + \varphi''(-2h) = (2 - \gamma) \varphi''(0) - 16 \sinh h A. \quad (15)$$

Step 3: elimination. Write $a_n = \varphi^{(n)}(0)/n!$. Taylor’s theorem with Lagrange remainders gives the following:

- $E_k = 2a_2(kh)^2 + 2a_4(kh)^4 + r_k$ with $|r_k| \leq 2(kh)^6 M_6/720$, for $k = 1, 2$;

- $\varphi''(2h) + \varphi''(-2h) = 4a_2 + 96a_4h^2 + r''$ with $|r''| \leq \frac{4}{3}h^4M_6$.

The two relations (15) determine a_2h^2 and a_4h^4 . Substituting them into E_1 gives, with $D = 4 - \gamma/3$, the exact identity

$$D E_1 = 4h^2 \sinh h A + \frac{1}{4}h^2 r'' - \left(\frac{\gamma}{16} + \frac{D}{4}\right)r_2 + D r_1.$$

Since $0 < \gamma < 12$, we have $0 < D < 4$, and the remainder is at most $C_6 h^6 M_6$ with $C_6 = \frac{\gamma}{90} + \frac{1}{3} + 4\left(\frac{2}{45} + \frac{1}{360}\right)$.

Step 4: the slope. By (15), $\varphi(2h) - \varphi(-2h) = -2\varphi(-2h) = -2e^h \cos(qZ; q^2)$. At a pole, $\cos(qZ; q^2) = qZ \sin(Z; q^2)$ and $|\sin(Z; q^2)| \geq \sqrt{1 - 0.61Z}$ for $Z \leq \min(0.2, q/3)$; both are proved in [5, proof of Proposition prop:selower], and here $Z \leq 0.1002$. Since $e^h qZ = z_0$, and Taylor's theorem gives $|\varphi(2h) - \varphi(-2h) - 4hA| \leq \frac{8}{3}h^3 M_3$, we get

$$|A| \geq \frac{z_0 |\sin(Z; q^2)|}{\tau} - \frac{\tau^2}{6} M_3 \geq w \left[\sqrt{\frac{1-q}{\tau}} \sqrt{1 - 0.61Z} - \frac{\tau}{3} \frac{M_3}{w^3} \right].$$

Step 5: derivative bounds. By [5, Lemma app:rep2] at radius $\rho = z_0 e^t$, $\psi(t) = \text{Re } \mathbb{E}[F]$. Here $F = (\rho e^{i\theta}; q)_{\infty}^{-1}$, $\theta = \frac{\pi}{2} + \sigma g$, $\sigma^2 = \tau/2$ and g is standard normal. We have $\partial_t F = -i\partial_{\theta} F$. Gaussian integration by parts, $\mathbb{E}[f^{(k)}(g)] = \mathbb{E}[f(g) \text{He}_k(g)]$ with He_k the Hermite polynomials, then gives

$$\psi^{(k)}(t) = \text{Re}[(-i)^k w^k \mathbb{E}[F \text{He}_k(g)]].$$

In $\log F = \sum_{n \geq 1} (\rho e^{i\theta})^n / (n(1 - q^n))$, the real parts are bounded term by term:

- $n = 1$: $-\rho \sin(\sigma g) / (1 - q) \leq \lambda |g|$, with $\lambda = \rho \sigma / (1 - q)$;
- $n = 2$: $-\kappa_2 \cos(2\sigma g) \leq -\kappa_2 + 2\kappa_2 \sigma^2 g^2$, with $\kappa_2 = \rho^2 / (2(1 - q^2))$, using $\cos 2u \geq 1 - 2u^2$;
- $n \geq 3$: at most $\rho^3 / (9q^2(1 - q)(1 - \rho)) \leq K_* := \rho^3 / (9q^2(1 - q)(1 - \rho/q))$, since $n(1 - q^n) \geq 3(1 - q^3) \geq 9q^2(1 - q)$.

Hence $|\psi^{(k)}(t)| \leq w^k e^{K_* - \kappa_2} \mathbb{E}[e^{\lambda|g| + \kappa_2 \tau g^2} |\text{He}_k(g)|]$. From $\varphi^{(k)} = e^{-t/2} \sum_j \binom{k}{j} (-\frac{1}{2})^{k-j} \psi^{(j)}$ and $1/(2w) = \sigma/2$, the same bound holds for $\varphi^{(k)}$ with the extra factor $e^{\tau/2}$, and with

$$\sum_j \binom{k}{j} (\sigma/2)^{k-j} |\text{He}_j(g)|$$

in place of $|\text{He}_k(g)|$. For $|t| \leq \tau$ and $\tau \leq \bar{\tau} = 5 \cdot 10^{-3}$, the parameters $\lambda, \rho, \sigma, \kappa_2 \tau, K_*$ are at most their values at $t = \tau = \bar{\tau}$. They are increasing in τ and t , and $\kappa_2 \geq e^{-2\bar{\tau}}/2$. Directed-rounding interval arithmetic evaluates the resulting Gaussian integrals on 6000 cells of $g \in [0, 14]$. The integrand is bounded beyond $g = 14$ by its value at 14, because it is decreasing there even after multiplication by e^g . The result is

$$\max_{|t| \leq \tau} |\psi''| \leq 3.24 w^2, \quad M_3 \leq 5.68 w^3, \quad M_6 \leq 53.3 w^6$$

(code/zeta_probe/nondfinite/vminusx_certificate.py, mpmath iv at 50 digits; the computed enclosures are 3.2196, 5.6320 and 52.193, rounded up to the stated constants, which are the ones used below).

Step 6: assembly. Each quantity below is monotone in τ , so it is evaluated at $\bar{\tau}$ with the rounded constants. Step 4 gives $|A| \geq 0.958 w$. By Taylor's theorem, $S_e = \psi(-h) = -hA(1 + \delta_2)$ with $|\delta_2| \leq hK_2 w / (2 \cdot 0.958) \leq 1.20\sqrt{\tau}$, where $K_2 = 3.24$. Step 3 and $\sinh h \geq h$ give $E_1 = (4h^2 \sinh h/D) A(1 + \delta_1)$ with $|\delta_1| \leq C_6 \cdot 53.3 \sqrt{\tau} / (4\sqrt{2} \cdot 0.958) \leq 5.14\sqrt{\tau}$. By (13),

$$\beta_m = \frac{e^{h/2} 4h^2 \sinh h}{2hD} \cdot \frac{1 + \delta_1}{1 + \delta_2} = \Lambda \tau^2 \frac{1 + \delta_1}{1 + \delta_2},$$

which is the two-sided bound. Finally, $\frac{1}{8} < \Lambda \leq \Lambda(\bar{\tau}) \leq 0.12527$, since $e^{h/2}$, $\sinh h/h$ and $1/D$ are increasing, and $\sqrt{\bar{\tau}} = 0.0707$. This gives $0.073\tau^2 \leq \beta_m \leq 0.187\tau^2$. In particular $\beta_m > 0$. $\square$

The six exceptional values and the uniform-range bracket are checked against direct evaluation at the travel poles in `code/zeta_probe/nondfinite/vminusx_check.py`, in floating point at 45 digits. Along the ladder, $\beta_m/\tau_m^2 \rightarrow \frac{1}{8}$: the value is 0.12560 at $m = 7$ and 0.12507 at $m = 19$, and $\beta_m/(\Lambda\tau_m^2) - 1$ is below $3 \cdot 10^{-3}$ there, against the certified 0.49 at $\bar{\tau}$. The error terms are lossy by a factor of about 10^2 ; the leading constant is not. The identity $\cos(qZ; q^2) = qZ \sin(Z; q^2)$ is checked there, and (14) in `vminusx_diffeq_check.py`, both to working precision. Only the uniform range is needed for infinitely many poles of V at $-x_m$.

6.2 D -finiteness

Lemma 25. *Let $\Omega \subset \mathbb{C}$ be a domain, let $a_0, \dots, a_r, b$ be holomorphic on Ω with $a_r \neq 0$, and let f be meromorphic on Ω with $a_r f^{(r)} + \dots + a_1 f' + a_0 f = b$ off the poles of f . Then every pole of f in Ω is a zero of a_r .*

Proof. Let z_0 be a pole of f of order $k \geq 1$. Then $f^{(j)}$ has a pole of order exactly $k + j$ at z_0 , so for $j < r$ the term $a_j f^{(j)}$ has a pole of order at most $k + r - 1$, and b is holomorphic. If $a_r(z_0) \neq 0$, the term $a_r f^{(r)}$ has a pole of order exactly $k + r$, which nothing else in the equation cancels. Hence $a_r(z_0) = 0$. $\square$

Theorem 26. *U and V are not D -finite: neither satisfies a linear differential equation*

$$\sum_{j=0}^r a_j(x) f^{(j)}(x) = b(x)$$

with polynomial coefficients and $a_r \neq 0$, nor even one whose coefficients a_j and b are holomorphic on a neighbourhood of the closed disc $|x| \leq 1$. Equivalently, (u_n) and (v_n) are not P -recursive. The same holds for the travel resolvent G_0^T , as a series in q and as a series in x , and for G_0^T it is unconditional. (Standing for U and V : Theorems 12 and 15.)

Proof. Let f be one of the series and suppose it satisfies such an equation. By Lemma 13 (for G_0^T , [5, Theorem thm:blocks], first clause) f continues to a meromorphic function on the open unit disc D . The expression $\sum_j a_j f^{(j)} - b$ is then meromorphic on D and vanishes near 0, so by the identity theorem it vanishes on D off the poles of f . By Lemma 25 every pole of f in D is a zero of a_r . Since $a_r \neq 0$ is holomorphic on a neighbourhood of the compact closed disc, its zeros there are isolated and hence finite in number. But f has infinitely many poles in D : V at every x_m (Theorem 18) and every $-x_m$ (Proposition 24), U at every $\pm x_m$ (Theorem 20 and Proposition 23), and G_0^T at every q_m in the variable q and at every $\pm x_m$ in the variable x (Theorem 17). This is a contradiction. D -finiteness of a power series is equivalent to P -recursiveness of its coefficients [28, §6.4]. $\square$

The pole argument does not reach differential algebraicity: $1/\cos(\pi/(1-x))$ has poles at $x = 1 - 2/(2n+1)$, $n \geq 1$, infinitely many in $|x| < 1$ and accumulating at $x = 1$, and satisfies an algebraic differential equation. Whether U and V are D -algebraic is open; we expect not.

Corollary 27. *The bivariate series $\mathcal{W}(x, y) \in \mathbb{Z}[y][[x]] \subset \mathbb{Z}[[x, y]]$ is not D -finite in (x, y) . (Standing: Theorems 12 and 15.)*

Proof. D -finiteness is preserved under substitution of algebraic power series without constant term [19]. If $\mathcal{W}$ were D -finite, the substitution $y = x^2$ would make $\mathcal{W}(x, x^2) = U$ D -finite ([5, Corollary cor:M3prime-growth]), contrary to Theorem 26. $\square$

6.3 Linear q -difference and Mahler equations

Proposition 28. *Let f be U or V , as series in x , or G_0^T , as a series in q , and let $a_0, \dots, a_r, b$ be polynomials with $a_0 a_r \neq 0$.*

(i) *If $Q \in \mathbb{C}^\times$ is not a root of unity, then $\sum_{i=0}^r a_i(x) f(Q^i x) \neq b(x)$.*

(ii) *If $k \geq 2$ is an integer, then $\sum_{i=0}^r a_i(x) f(x^{k^i}) \neq b(x)$.*

For G_0^T both statements are unconditional ((ii) uses the window statement of [5, Lemma lem:infpoles], hence the verified bound of [5, Lemma lem:T2abs]); for U and V the standing is that of Theorems 18 and 20. When Q is a root of unity the case is open: the pole set can be invariant under $x \mapsto Qx$, as that of U is under $x \mapsto -x$ (Proposition 23), and the argument below gives nothing.

Proof. Let D be the open unit disc, P the pole set of f in D , and $\rho > 0$ the radius of convergence of f ; P lies in $\rho \leq |x| < 1$, has finitely many points in each disc $|x| \leq R < 1$, and is infinite (proof of Theorem 26). An equation as in (i) or (ii) holds as an identity of power series near 0.

(i) If $|Q| > 1$, replacing x by $Q^{-r}x$ and i by $r - i$ gives an equation of the same form with Q^{-1} in place of Q and new leading and trailing coefficients $a_r(Q^{-r}x)$, $a_0(Q^{-r}x)$; so we may assume $|Q| \leq 1$. Then every $f(Q^i x)$ is meromorphic on D , and the equation holds on D by the identity theorem. Let E be the finite zero set of a_0 in D . At $p \in P \setminus E$ the term $a_0 f$ is singular and b is not, so some $f(Q^i x)$ with $1 \leq i \leq r$ is singular at p , that is $Q^i p \in P$. Thus every $p \in P \setminus E$ has a successor $Q^i p \in P$, $1 \leq i \leq r$, and every point of P is the successor of at most r points. Follow successors from any $p \in P$; along the chain the exponent of Q increases strictly.

If $|Q| < 1$ the moduli decrease strictly, so the chain stays in the finite set $P \cap \{|x| \leq |p|\}$ and ends, necessarily at a point of E . A chain of n steps ends at modulus at most $|Q|^n |p| < |Q|^n$, and that modulus is at least ρ , so $n < N := \log \rho / \log |Q|$. Hence every point of P is reached backwards from E in fewer than N steps, and $|P| \leq |E| \sum_{n < N} r^n$ is finite.

If $|Q| = 1$ the chain stays on the circle $|x| = |p|$, which meets P in a finite set, and its points are pairwise distinct because Q is not a root of unity; so it ends in E . Hence P lies on the finitely many circles $|x| = |e|$, $e \in E$, each meeting P in finitely many points, and P is finite. Both cases contradict the infinitude of P .

(ii) Here every $f(x^{k^i})$ is meromorphic on D and the equation holds on D . Work on the positive axis: let $P_+ = P \cap (0, 1)$ and let E_+ be the finite zero set of a_0 in $(0, 1)$. As in (i), every $p \in P_+ \setminus E_+$ has a successor $p^{k^i} \in P_+$ with $1 \leq i \leq r$; it is smaller than p , so every chain ends in E_+ , and reading a chain backwards, $p = e^{k^{-n}}$ for some $e \in E_+$ and some integer $n \geq 0$. Hence, for $0 < R < 1$ and $\ell = \log(1/R)$,

$$\#(P_+ \cap (0, R]) \leq \sum_{e \in E_+} \#\{n \geq 0 : k^n \leq \log(1/e)/\ell\} \leq |E_+| \left(1 + \log_k \frac{L}{\ell}\right), \quad L = \max_{e \in E_+} \log(1/e),$$

which grows like $\log(1/\ell)$ as $R \uparrow 1$. On the other hand, by [5, Lemma “infinitude and accumulation of the travel poles”, lem:infpoles] there is a travel pole with $w \in (m\pi, (m+1)\pi)$ for every $m \geq 4$, at $x = e^{-1/w^2} < e^{-1/((m+1)\pi)^2}$, equivalently $q = e^{-2/w^2} < e^{-2/((m+1)\pi)^2}$, and it lies in P_+ (Theorems 18, 20, 17). Distinct windows give distinct poles, so $P_+ \cap (0, R]$ has at least $\lfloor c/(\pi\sqrt{\ell}) \rfloor - 4$ points, with $c = 1$ in the variable x and $c = \sqrt{2}$ in q . This grows like $\ell^{-1/2}$, faster than $\log(1/\ell)$, a contradiction for R close to 1. $\square$

The count in (ii) is made on the real axis on purpose. It uses only the travel poles that are proved to exist, and it needs no statement about where on the circle the poles of f accumulate; compare Remark 31. Mahler equations are those of [20]. The catalytic equation of §2 is a q -difference equation in the catalytic variable, not in x , and is not affected by (i).

6.4 The natural boundary

We use the following rationality theorem.

Theorem 29 (Pólya–Bertrandias). *Let $\Omega \subset \mathbb{C}$ be a simply connected domain containing 0 and stable under complex conjugation, and let φ be the Riemann map of the unit disc onto Ω with $\varphi(0) = 0$, $\varphi'(0) > 0$. If $\varphi'(0) > 1$ and $f \in \mathbb{Z}[[x]]$ is the Taylor series at 0 of a function meromorphic on Ω , then f is a rational function.*

This is Pólya’s theorem [24], which replaced the disc of Borel’s rationality theorem by a domain measured by transfinite diameter; Bertrandias [2] extended it to p -adic and adelic domains, whence the name. Bost and Chambert-Loir prove a generalisation to curves over number fields [7, Thm. 7.9], from which they recover the Pólya–Bertrandias criterion [7, Example 7.10], and Cantor’s rationality theorem [9, Thm. 5.2.2] is a capacity-theoretic generalisation. Meromorphy suffices, but the strict inequality $\varphi'(0) > 1$ is needed: for Ω the unit disc, $\varphi'(0) = 1$, and $\sum_n x^{2^n}$ is not rational.

Theorem 30. (a) *The travel resolvent $G_0^T = \Sigma_0/(1-\Sigma_1) \in \mathbb{Z}[[q]]$ has $|q| = 1$ as a natural boundary. This is unconditional.*

(b) *U and V have $|x| = 1$ as a natural boundary. (Standing: Theorems 12 and 15.)*

More precisely, in each case there is no point ζ of the unit circle and no disc B about ζ such that the function continues meromorphically to $D \cup B$, D the open unit disc.

Proof. Let f be G_0^T (in q), U or V (in x). In each case f has integer coefficients (the travel-run counts, u_n, v_n), is meromorphic on D (Lemma 13), and is not rational, having infinitely many poles in D (Theorems 17, 18, 20). Suppose f continues meromorphically to $D \cup B$ with B the disc of radius $\delta > 0$ about ζ , $|\zeta| = 1$. Shrinking δ , we may assume $\delta < |\operatorname{Im} \zeta|$ when $\zeta \notin \mathbb{R}$, so that B and its conjugate $\bar{B}$ are disjoint; when $\zeta = \pm 1$, $\bar{B} = B$. Because the coefficients are real, $\overline{f(\bar{x})} = f(x)$ near 0, hence on D , and $x \mapsto \overline{f(\bar{x})}$ continues f meromorphically to $\bar{B}$, agreeing with f on $\bar{B} \cap D$. So f is meromorphic and single-valued on

$$\Omega = D \cup B \cup \bar{B},$$

which is stable under conjugation and simply connected: it is the disc D with one or two disjoint discs attached, each meeting D in a convex set. Let φ be its Riemann map, $\varphi(0) = 0$, $\varphi'(0) > 0$. By uniqueness of the Riemann map, $\overline{\varphi(\bar{z})} = \varphi(z)$, so φ has real Taylor coefficients (Schwarz reflection). The map $\psi = \varphi^{-1}|_D$ sends D into D with $\psi(0) = 0$ and is not onto, since $\Omega \neq D$; by the Schwarz lemma $0 < \psi'(0) < 1$, that is $\varphi'(0) > 1$. Theorem 29 now makes f rational, a contradiction. $\square$

A D -finite series continues analytically along every path avoiding the finitely many zeros of the leading coefficient of its equation, so Theorem 30 gives a second proof of the polynomial-coefficient case of Theorem 26. The first proof is elementary and also covers coefficients holomorphic near the closed disc.

Remark 31 (where on the circle the poles accumulate). Theorem 30 does not locate the singularities on the circle. The poles proved above accumulate only at $x = \pm 1$ (at $q = 1$ for G_0^T), yet every point of the circle is singular. Numerically the pole set is much larger. The function $1 - \Sigma_1 = \cos(Z; q^2)$ has 64 zeros in $|q| < 0.94$ by the argument principle: the travel poles q_1, q_2 and 62 non-real zeros in 31 conjugate pairs, the first at $q = -0.4075 \pm 0.5449i$. At each non-real zero Π_q was found non-zero at 40 digits, so these are candidate poles of G_0^T and of U off the real axis. Plotted, the zeros approach roots of unity along straight lines. We conjecture that the poles are dense on the circle; a route is a q -cosine resummation at $q = \zeta e^{-t}$, ζ a root of unity. The natural boundary does not depend on this.

7 Open questions

The travel resolvent $\Sigma_0/(1 - \Sigma_1)$ is transcendental, not D -finite, and has $|q| = 1$ as a natural boundary, unconditionally. The growth series U of W_{univ} and its relaxed companion V have the same properties in x , on the transfer model (Theorem 12) and the junction pairing (Theorem 15); two finite ranges are computer-assisted, the six gate poles above $\tau = 5 \cdot 10^{-3}$ and the positivity of the pairing at the first twelve poles, and neither is needed for transcendence, only for the statements “at every travel pole”. The series in question is the growth series of the universal group W_{univ} ; for a particular right triangle it agrees with that of W_T only to the effective universality depth of [3]. Several questions remain open.

- (1) Are the poles of U dense on the unit circle (Remark 31)?
- (2) Are U and V D -algebraic? The pole argument does not decide this.
- (3) Is the number β_2 transcendental? The question reduces to a value statement for a Hahn–Exton q -Bessel function, and the obstruction sits at the arithmetic ratio $\rho = \frac{1}{2}$ [5, Appendix app:beta2].
- (4) Is $(u_n \bmod p)$ non-automatic for some prime p ? By Christol’s theorem this would give a second, analysis-free proof of transcendence; the p -kernel is computed to be maximally non-automatic at $p = 3, 5$ [5, §sec:modp].
- (5) The number of travel poles in a window of w , equivalently an asymptotic count of the zeros of $\cos(\cdot; q^2)$ in $(0, Z(q))$ uniform as $q \rightarrow 1$, is not known [5, Remark “what the known theory does not give”, rem:knowntheory].

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